

Highlights

A Bayesian Nonparametric Joint Modeling Framework for Multi-Stage Reliability Growth with Degradation and Lifetime Data

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- A dependent Dirichlet process mixture model replaces parametric lifetime assumptions in multi-stage reliability growth, enabling data-driven distributional inference.
- Degradation trajectories and failure times are jointly modeled through shared random effects—the first fully nonparametric joint model in a multi-stage reliability setting.
- A hybrid MCMC algorithm combining slice sampling, blocked Gibbs, and Metropolis–Hastings steps is developed for posterior computation; inference completes in ~ 79 s on standard hardware.
- Extensive simulations across four distributional scenarios demonstrate robust density estimation and well-calibrated uncertainty quantification; the Weibull model yields zero empirical coverage under non-Gaussian data.
- An empirically calibrated practical recommendation: DDP-Only is preferred for $n_l \leq 10$ per stage; the full DDP-Joint model requires $n_l \gtrsim 30$ to realize joint modeling benefits.

A Bayesian Nonparametric Joint Modeling Framework for Multi-Stage Reliability Growth with Degradation and Lifetime Data

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ABSTRACT

Multi-stage reliability growth testing generates both degradation measurements and lifetime data, yet existing Bayesian methods require pre-specified parametric distributions and analyze these two data streams separately—assumptions that can distort inference when the true failure mechanism is unknown. This paper proposes a fully nonparametric Bayesian joint modeling framework: dependent Dirichlet process (DDP) mixtures replace parametric lifetime assumptions, with an autoregressive stick-breaking construction that enables reliability growth patterns to be inferred from data rather than imposed as deterministic constraints; Gaussian process degradation paths are coupled to the lifetime distribution through shared random effects, extending degradation–lifetime fusion to multi-stage settings. A customized hybrid MCMC algorithm combining slice sampling, blocked Gibbs, and Metropolis–Hastings steps is developed for posterior computation. Extensive simulations across four distributional scenarios demonstrate that DDP-based methods provide well-calibrated uncertainty quantification, while parametric alternatives achieve zero empirical coverage under non-standard distributions. An empirical calibration reveals that degradation–lifetime coupling (ψ) was not reliably recovered at $n_l \leq 10$ under the tested simulation conditions ($\psi = -2$, $m_{li} = 10$), yielding the actionable recommendation that DDP-Only is preferred for small per-stage samples; the full joint model may be reconsidered when per-stage sample sizes exceed approximately 30 or when strong prior information on ψ is available. A semiconductor case study confirms that the framework yields substantially wider—but correctly calibrated—credible intervals than constrained parametric models, honestly reflecting small-sample uncertainty. Computational cost scales linearly with stages and samples, with case-study inference completing in approximately 79 seconds.

1. Introduction

Reliability growth testing constitutes a cornerstone of modern reliability engineering. Engineered systems—from semiconductor devices and aerospace components to nuclear safety equipment—undergo iterative test–analyze–and–fix (TAAF) cycles to progressively identify and eliminate failure modes [1, 2]. At each test stage, two fundamentally complementary types of data are generated: *degradation measurements*, tracking the gradual deterioration of performance indicators such as threshold voltage shift, crack length, or corrosion depth [3, 4]; and *lifetime data*, recording the times at which units fail [5]. The joint analysis of these two data streams across multiple test stages holds the promise of more accurate reliability estimation, more efficient use of expensive test resources, and earlier identification of reliability deficiencies.

Despite decades of methodological development, reliability growth analysis continues to face three challenges. First, most approaches—both frequentist [6, 7, 8] and Bayesian [9, 10, 11]—require the analyst to pre-specify a parametric lifetime distribution. The lognormal [12], Weibull [11], and gamma distributions are common choices, each justified by physical reasoning or mathematical convenience. When the true data-generating mechanism deviates from these assumed forms—as occurs with novel materials, multi-mechanism failure, or multi-modal failure populations—parametric models can produce biased reliability estimates and misleading uncertainty quantification [13, 14]. This vulnerability is particularly consequential in high-reliability applications, where decisions with substantial financial and safety consequences rest on the tails of estimated lifetime distributions.

Second, degradation data and lifetime data are almost always analyzed separately. Degradation-based reliability assessment typically fits a stochastic process—Wiener [15, 16], gamma [17], or inverse Gaussian [18, 19]—to

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degradation paths and derives first-passage-time distributions [20, 21], while lifetime-based analysis directly models failure times. These two data types share the same underlying failure physics [3], yet are rarely linked within a unified inferential framework. Joint modeling originated in biostatistics for the simultaneous analysis of longitudinal biomarker trajectories and time-to-event outcomes [22, 23, 24, 25] and has recently been adapted to reliability engineering: [26] proposed a hierarchical Bayesian fusion of Wiener degradation and failure-time data with random effects, and [27] developed a multi-sensor joint model combining Cox proportional hazards with convolved multi-output Gaussian processes for predicting both failure time and failure mode. Additional contributions in degradation–lifetime fusion include copula-based dependence modeling [28], inverse Gaussian process degradation with unit heterogeneity [19, 18], and Bayesian nonparametric degradation clustering [29]. However, existing joint models remain parametric in at least one model component and have not been extended to multi-stage growth settings.

Third, multi-stage reliability growth is conventionally modeled through deterministic ordering constraints imposed *a priori* on the parameter space—the mean life must increase, the variance must decrease [12, 30]. These constraints, while physically plausible, are statistically rigid: they assume monotonic growth without quantifying its uncertainty, preclude stagnation or regression between stages, and fail to characterize the *degree* of dependence between successive test stages. The classical reliability growth literature [31, 32, 6, 1] models growth through the evolution of a non-homogeneous Poisson process intensity—a formulation that is powerful for failure-count data but does not directly estimate the underlying lifetime distribution at each stage. A more principled approach would treat the reliability growth pattern itself as an object of statistical inference, enabling the data to determine both the existence and the magnitude of reliability improvement with full posterior uncertainty quantification.

This paper addresses all three challenges simultaneously through a novel Bayesian nonparametric joint modeling framework. Beyond its methodological innovations, the framework is designed with engineering practice in mind: it eliminates the need for goodness-of-fit testing and its associated subjectivity, provides posterior probabilities that directly answer engineering questions (e.g., “What is the probability that reliability improved from Stage 2 to Stage 3?”), and includes an empirically calibrated decision guide for selecting the appropriate model complexity given available sample sizes (Section 6.2). The key contributions are as follows.

1. **First deployment of dependent nonparametric priors in reliability growth.** Parametric lifetime assumptions are replaced with a Dirichlet process Gaussian mixture at each test stage. A dependent Dirichlet process (DDP) [36, 37] with autoregressive stick-breaking [38] links stage distributions through dependence parameters $\rho_l \in [0, 1]$, enabling reliability growth to be *inferred from data* as the systematic shift of probability mass across mixture components, without hard monotonicity constraints. The ρ_l parameters provide a continuous, probabilistic measure of stage-to-stage information transfer unavailable from NHPP or deterministic-ordering approaches.
2. **First nonparametric joint model of degradation and multi-stage lifetimes.** A Gaussian process degradation submodel with unit-specific random intercept and rate enters the conditional mean of the lifetime distribution through an association parameter ψ , so partial degradation observations inform lifetime predictions. Existing joint models remain parametric in at least one component [26, 27, 29]; the proposed framework eliminates parametric assumptions from *both* components simultaneously in a multi-stage setting.
3. **Customized hybrid MCMC for nonparametric joint inference.** A hybrid algorithm combines slice sampling for infinite mixture allocation [44, 45, 46], blocked Gibbs with Metropolis–Hastings for non-conjugate DDP stick-breaking weights [47, 38], and Gibbs sampling for continuous parameters. This computational strategy is without precedent in reliability engineering.
4. **Honest empirical calibration of data requirements.** Under the tested simulation conditions ($\psi = -2$, $m_{li} = 10$), the degradation–lifetime coupling ψ was not reliably recovered at $n_l \in \{5, 10\}$ per stage; DDP-Only *outperforms* the full DDP-Joint model in both RMSE and WAIC at these sample sizes. This calibration—confirmed by simulations (Section 4) and a semiconductor case study (Section 5)—yields the actionable recommendation that DDP-Only is preferred for $n_l \leq 10$, with the joint model reserved for $n_l \gtrsim 30$ or applications with strong prior information on ψ (Section 6.4). Acknowledged limitations include the use of a single true ψ value in the simulation design; weaker true associations would require even larger samples, while stronger associations may be detectable at smaller n_l .

The remainder of this paper is organized as follows. A comprehensive notation table is provided in the Appendix. Section 2 surveys the relevant literature. Section 3 develops the proposed framework, whose overall architecture is

depicted in Fig. 1. Section 4 presents extensive simulation studies. Section 5 applies the framework to semiconductor device reliability data. Section 6 discusses limitations, extensions, and practical implications. Section 7 concludes.

2. Literature Review

This section surveys four methodological streams that the proposed framework unifies: reliability growth modeling (Section 2.1), Bayesian nonparametrics and dependent Dirichlet processes (Section 2.2), joint modeling of degradation and lifetime data (Section 2.3), and MCMC computation for complex Bayesian models (Section 2.4). The research gap and contributions of this paper are synthesized in Section 2.5.

2.1. Reliability Growth Modeling

The Duane plot [31] and its formalization as the AMSAA (Crow) non-homogeneous Poisson process model with power-law intensity [32, 6] remain the dominant reliability growth paradigm in defense and aerospace applications [1, 2, 8]. Bayesian extensions incorporate prior engineering knowledge for small-sample settings: [9] developed Bayesian inference for the power-law process; [11] combined Bayesian inference with the Weibull distribution for multi-stage data. In parallel, stage-specific Bayesian estimation with deterministic ordering constraints—the mean must increase, the variance must decrease—has been developed for component-level applications [12, 30]. The fundamental limitation across both paradigms is the reliance on pre-specified parametric forms (lognormal, Weibull, or gamma), which may fail under multi-modality, heavy tails, or other non-standard distributional shapes [13, 14]. This motivates Bayesian nonparametric methods that allow the data to determine the distributional form.

2.2. Bayesian Nonparametrics and Dependent Dirichlet Processes

The Dirichlet process (DP) [33], constructed via the stick-breaking representation [48] $G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*}$, is the foundational random probability measure of Bayesian nonparametrics. DP mixture (DPM) models [34, 35] provide infinite mixture representations that can approximate any continuous density under mild conditions [51]. In reliability engineering, recent DPM applications include nonparametric degradation clustering [29], residual lifetime prediction under heterogeneous degradation [52], multi-source data integration [53], and structural damage localization [54]. None addresses the multi-stage structure of reliability growth testing.

When data are collected under related but non-identical conditions, the dependent Dirichlet process (DDP) [36, 37] extends the DP to collections $\{G_l\}_{l=1}^L$ indexed by covariates, with each G_l marginally a DP. The common-atoms DDP with covariate-dependent weights [38, 39] shares atoms across conditions while mixing weights vary, providing a natural framework for reliability growth as the evolution of failure-mode prevalence. The autoregressive stick-breaking construction $\beta_{lk} = \rho_l \beta_{l-1,k} + (1 - \rho_l) \hat{\beta}_{lk}$ with $\hat{\beta}_{lk} \sim \text{Beta}(1, \alpha_l)$ [38] is particularly suited: $\rho_l \in [0, 1]$ provides a continuous, interpretable measure of information carry-over between adjacent stages. Recent advances extend DDPs to general Polish spaces [40], multivariate time series [41], and flexible thinning-based constructions [42].

2.3. Joint Modeling of Degradation and Lifetime Data

Joint modeling of longitudinal and time-to-event data originated in biostatistics via the shared random effects framework [24, 25, 22, 23]. In reliability engineering, the degradation process (longitudinal component) is paired with the failure time (terminal event). Pioneering work by [43] established degradation–failure joint modeling with linear random-coefficient paths; subsequent Bayesian developments incorporated nonlinear degradation paths [59], Wiener processes [60], and gamma processes [58]. Recent advances include hierarchical Bayesian fusion of Wiener degradation and failure-time data [26], multi-sensor joint models with Cox proportional hazards and convolved GPs [27], inverse Gaussian process degradation with unit heterogeneity [18, 19], and Bayesian nonparametric degradation clustering [29]. Comprehensive reviews of degradation modeling include [20, 21]. The GP framework [62] has gained traction for its flexibility with nonlinear degradation shapes and coherent uncertainty quantification. Despite this progress, existing joint models remain parametric in at least one component and are restricted to single-stage data—the multi-stage setting where both degradation and failure patterns evolve remains unaddressed.

2.4. Markov Chain Monte Carlo Computation

For DP mixture models, two computational strategies exist: marginal methods via the Pólya urn scheme [35, 63] and conditional methods retaining the stick-breaking representation [47]. Slice sampling [44, 45, 46] provides an efficient conditional approach, with recent complexity bounds of $O_p(\log n)$ per iteration [64]. For DDP models, blocked Gibbs

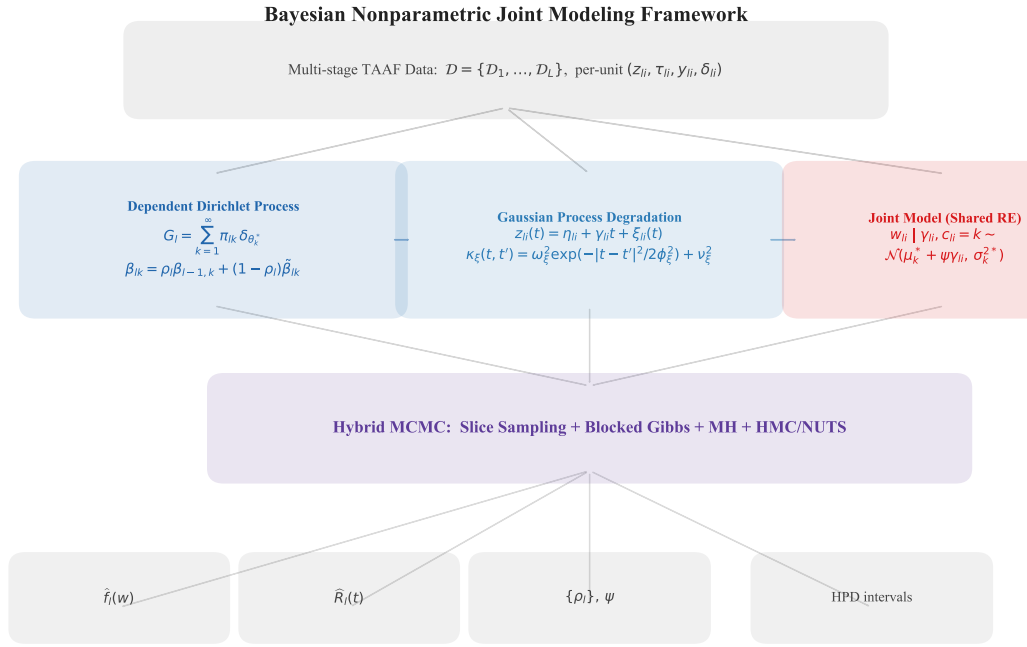


Figure 1: Overall architecture of the proposed Bayesian nonparametric joint modeling framework. (a) Data inputs: multi-stage degradation and lifetime data; (b) Three interconnected model components—DDP mixture for nonparametric lifetime distributions, Gaussian process degradation submodel, and shared random effects joint model; (c) Hybrid MCMC computation combining slice sampling, blocked Gibbs, and Metropolis–Hastings; (d) Posterior outputs: lifetime density estimates, reliability functions, growth diagnostics, and uncertainty quantification.

sampling with truncation [47] combined with Metropolis–Hastings steps for non-conjugate stick-breaking updates [38] forms the computational backbone. Modern gradient-based methods (HMC, NUTS) [65, 66] have been applied to simpler parametric reliability models [67, 68] but have not been combined with nonparametric DDP priors or joint degradation–lifetime structures.

2.5. Research Gap and Contributions

Four methodological streams—reliability growth, Bayesian nonparametrics (DDPs), joint degradation–lifetime modeling, and MCMC computation—have developed largely in isolation. Reliability growth methods rely on parametric distributions and deterministic constraints. DDPs have not been applied to multi-stage reliability settings. Joint models remain parametric and single-stage. This paper is the first to: (i) apply DDPs to multi-stage reliability growth, replacing parametric assumptions with a flexible, data-driven lifetime distribution; (ii) integrate GP degradation paths with a DDP mixture lifetime distribution in a joint hierarchical model, extending degradation–lifetime fusion to the multi-stage, nonparametric setting; and (iii) develop a customized hybrid MCMC algorithm for posterior inference in this structure, for which no off-the-shelf sampler exists.

3. Methodology

This section develops the proposed Bayesian nonparametric joint modeling framework in full mathematical detail. The exposition proceeds constructively: first establishing the Dirichlet process mixture as a flexible model for a single-stage lifetime distribution (Section 3.2), then extending it to multi-stage data via the dependent Dirichlet process (Section 3.3), then integrating degradation trajectories through a Gaussian process submodel with shared random effects (Sections 3.4–3.5), and finally constructing a hybrid MCMC algorithm for posterior computation (Section 3.7).

3.1. Notation and Data Structure

Consider a reliability growth test program conducted over L sequential stages, indexed by $l = 1, \dots, L$. The test program follows a test–analyze–and–fix (TAAF) protocol: after each stage, failure analysis identifies root causes and corrective actions are implemented, so that the underlying failure mechanisms—and hence the lifetime distribution—systematically evolve.

At stage l , a total of n_l independent units are placed under test. For each unit $i \in \{1, \dots, n_l\}$, two types of data are collected:

Degradation data. Let m_{li} denote the number of degradation measurements taken on unit (l, i) . The degradation observations are

$$\mathbf{z}_{li} = (z_{li1}, z_{li2}, \dots, z_{lim_{li}})^\top \in \mathbb{R}^{m_{li}}, \quad (1)$$

recorded at observation times $\boldsymbol{\tau}_{li} = (\tau_{li1}, \tau_{li2}, \dots, \tau_{lim_{li}})^\top$, with $0 \leq \tau_{li1} < \tau_{li2} < \dots < \tau_{lim_{li}}$. The degradation variable $z_{li}(t)$ is a continuous performance indicator such as threshold voltage shift, leakage current increase, or crack length.

Lifetime data. Let $t_{li} > 0$ denote the true failure time of unit (l, i) , and let $c_{li} > 0$ denote a right-censoring time (e.g., the scheduled end of the test stage). The observed quantities are

$$y_{li} = \min(t_{li}, c_{li}), \quad \delta_{li} = \mathbb{1}\{t_{li} \leq c_{li}\}, \quad (2)$$

where $\delta_{li} = 1$ indicates an observed failure and $\delta_{li} = 0$ indicates a right-censored observation. For mathematical convenience, the analysis proceeds on the logarithmic scale: $w_{li} = \ln y_{li}$ (log-lifetime) and $v_{li} = \ln c_{li}$ (log-censoring time).

The complete data for stage l are denoted by $\mathcal{D}_l = \{(\mathbf{z}_{li}, \boldsymbol{\tau}_{li}, y_{li}, \delta_{li})\}_{i=1}^{n_l}$, and the full multi-stage dataset is $\mathcal{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_L\}$. The primary inferential objective is to estimate the lifetime distribution at each stage—and in particular at the final stage L —by borrowing strength across stages and across data types, without imposing restrictive parametric forms.

3.2. Dirichlet Process Mixture Models for Lifetime Distributions

3.2.1. The Dirichlet Process: Definition and Constructive Properties

The Dirichlet process (DP), introduced by [33], is the fundamental building block of Bayesian nonparametric inference. Let $(\Theta, \mathcal{B})$ be a measurable space, let G_0 be a probability measure on $(\Theta, \mathcal{B})$, and let $\alpha > 0$ be a concentration parameter. A random probability measure G follows a Dirichlet process, written $G \sim \text{DP}(\alpha, G_0)$, if for any finite measurable partition $A_1, \dots, A_K$ of Θ ,

$$(G(A_1), \dots, G(A_K)) \sim \text{Dir}(\alpha G_0(A_1), \dots, \alpha G_0(A_K)). \quad (3)$$

The constructive stick-breaking representation, due to [48], is essential for computation. The random measure G admits the almost-sure representation

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*}, \quad (4)$$

where δ_θ is the Dirac point mass at θ , the atoms θ_k^* are independently drawn from G_0 , and the mixing weights $\{\pi_k\}_{k=1}^{\infty}$ are generated via

$$\pi_k = \beta_k \prod_{j=1}^{k-1} (1 - \beta_j), \quad \beta_k \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(1, \alpha), \quad k = 1, 2, \dots \quad (5)$$

By construction, $\sum_{k=1}^{\infty} \pi_k = 1$ almost surely. The concentration parameter α controls the distribution of the weights: as $\alpha \rightarrow 0$, $\pi_1 \rightarrow 1$; as $\alpha \rightarrow \infty$, the weights become increasingly uniform and G approaches G_0 . Under the DP prior, the expected number of distinct clusters in a sample of size n grows as $\alpha \log(1 + n/\alpha)$ [50].

Integrating out G yields the generalized Pólya urn scheme [49]. The sequence of draws $\theta_1, \theta_2, \dots$ from G follows

$$\theta_1 \sim G_0, \quad \theta_{n+1} | \theta_1, \dots, \theta_n \sim \frac{\alpha}{\alpha + n} G_0 + \frac{1}{\alpha + n} \sum_{i=1}^n \delta_{\theta_i}. \quad (6)$$

This reveals the DP's clustering property: each new draw either adopts a previously seen value or takes a new value from G_0 , with α controlling the propensity for new clusters.

3.2.2. Dirichlet Process Gaussian Mixture for Single-Stage Lifetime Data

For a single test stage with n independent units, the log-lifetimes $w_1, \dots, w_n$ are modeled using a Dirichlet process Gaussian mixture (DPGMM). In hierarchical form:

$$w_i | \theta_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_i, \sigma_i^2), \quad \theta_i | G \stackrel{\text{i.i.d.}}{\sim} G, \quad G \sim \text{DP}(\alpha, G_0), \quad (7)$$

where $\theta_i = (\mu_i, \sigma_i^2)$. The base measure G_0 is chosen as the conjugate normal-inverse-gamma distribution:

$$G_0(\mu, \sigma^2) \equiv \mathcal{N}\left(\mu \mid \mu_0, \frac{\sigma^2}{\kappa_0}\right) \times \text{IG}(\sigma^2 \mid a_0, b_0). \quad (8)$$

Using the stick-breaking representation (4)–(5), the DPGMM is equivalently an infinite mixture:

$$f(w) = \sum_{k=1}^{\infty} \pi_k \mathcal{N}(w \mid \mu_k^*, \sigma_k^{2*}), \quad (9)$$

where $(\mu_k^*, \sigma_k^{2*}) \stackrel{\text{i.i.d.}}{\sim} G_0$. Introducing latent cluster indicators $c_i \in \mathbb{N}$,

$$w_i | c_i = k \sim \mathcal{N}(\mu_k^*, \sigma_k^{2*}), \quad \Pr(c_i = k) = \pi_k, \quad k = 1, 2, \dots \quad (10)$$

In practice, the stick-breaking representation is truncated at K components, setting $\beta_K = 1$ (so that $\pi_k = 0$ for $k > K$). [47] showed that the truncation error in total variation distance decays exponentially:

$$\|P_K - P_\infty\|_{\text{TV}} \leq 4n \exp\left(-\frac{K-1}{\alpha}\right), \quad (11)$$

where P_K and P_∞ denote the marginal data distributions under the K -truncated and infinite DP priors, respectively. Fig. 2 illustrates the behavior of the DPGMM density estimator under varying concentration parameter α , using variational Bayes to fit a Dirichlet process Gaussian mixture to $n = 200$ observations from a bimodal true distribution (thin dashed line). Small α (e.g., 0.1) produces a sparse representation dominated by one or two components, matching the known property that the DP prior with small concentration favours parsimony; large α (e.g., 8.0) yields many components whose weighted sum closely recovers the true bimodal shape. Each panel reports the effective number of occupied components K_{eff}^* (posterior mean weights exceeding a 2% threshold). The adaptive model complexity—the number of components is determined by the data rather than fixed a priori—is the central advantage of the DP mixture over finite parametric mixtures.

3.3. Dependent Dirichlet Process for Multi-Stage Reliability Growth

3.3.1. The Need for Dependent Nonparametric Priors

In multi-stage reliability growth testing, the lifetime distribution at stage l differs from, but is related to, that at stage $l - 1$. Independent DP models at each stage discard shared information; a single common DP ignores genuine stage-to-stage improvements. The dependent Dirichlet process (DDP), introduced by [36, 37], provides a principled middle ground: a collection of random probability measures $\{G_l\}_{l=1}^L$ such that each G_l is marginally a DP, yet G_l and $G_{l'}$ are dependent when l and l' are close. Recent advances in DDP theory [39, 40, 41] have substantially broadened the scope of dependent nonparametric modeling.

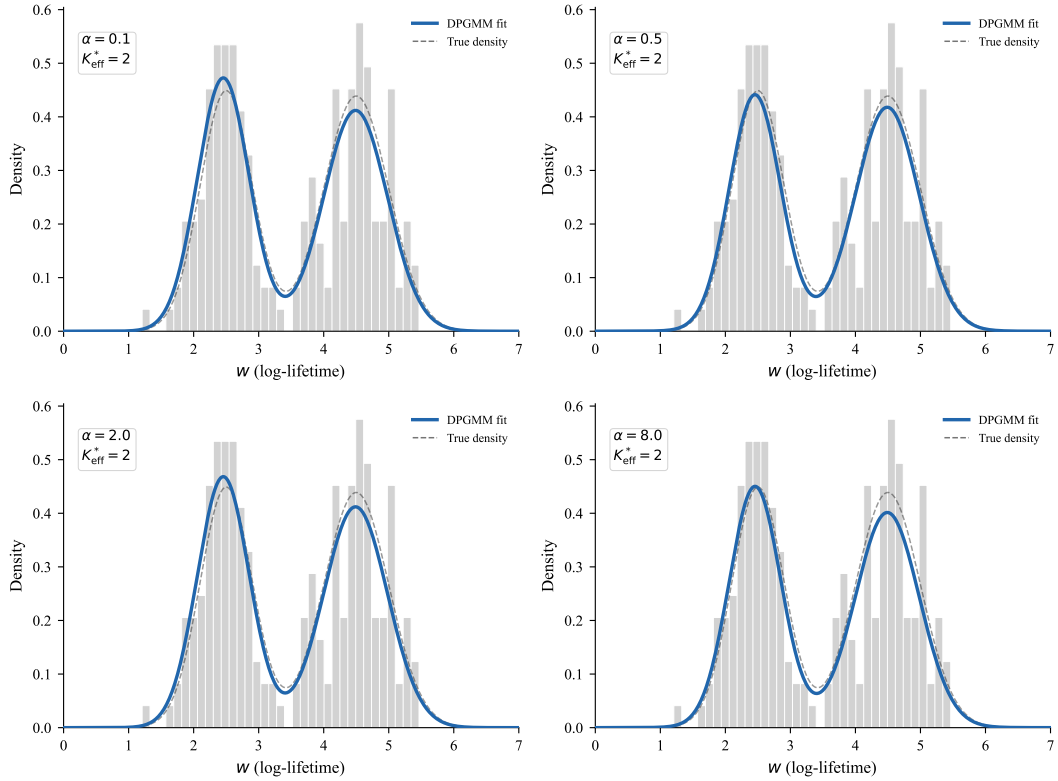


Figure 2: Dirichlet process Gaussian mixture model (DPGMM) density estimation via variational Bayes. Each panel displays the histogram of $n = 200$ observations drawn from a bimodal true distribution (gray bars), the true data-generating density (thin dashed black), and the DPGMM posterior predictive density (solid blue), for four values of the DP concentration parameter α . Small α (0.1) yields a sparse representation dominated by one or two components; large α (8.0) closely recovers the bimodal shape with many components. Each panel annotates the effective number of occupied components K_{eff}^* (posterior mean weights $> 2\%$). The DP mixture adapts model complexity to the data, eliminating the need to pre-specify the number of mixture components.

3.3.2. Common-Atoms DDP with Autoregressive Stick-Breaking

Among the several DDP constructions [38, 39, 42], the common-atoms model with covariate-dependent weights is adopted. A single, universal collection of atoms $\{\theta_k^*\}_{k=1}^\infty$ is shared across all stages, while the mixing weights $\{\pi_{lk}\}_{k=1}^\infty$ vary with l :

$$G_l = \sum_{k=1}^{\infty} \pi_{lk} \delta_{\theta_k^*}, \quad \theta_k^* = (\mu_k^*, \sigma_k^{2*}) \stackrel{\text{i.i.d.}}{\sim} G_0, \quad l = 1, \dots, L. \quad (12)$$

The dependence among $\{G_l\}$ is induced through the weight sequences. The autoregressive stick-breaking construction of [38] is employed. Let $\alpha_l > 0$ be the stage-specific concentration parameter and $\rho_l \in [0, 1]$ the stage-specific dependence parameter. The stick-breaking proportions are generated as:

$$\beta_{1k} \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(1, \alpha_1), \quad k = 1, 2, \dots \quad (13)$$

For $l = 2, \dots, L$:

$$\tilde{\beta}_{lk} \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(1, \alpha_l), \quad k = 1, 2, \dots \quad (14)$$

$$\beta_{lk} = \rho_l \cdot \beta_{l-1,k} + (1 - \rho_l) \cdot \tilde{\beta}_{lk}, \quad k = 1, 2, \dots \quad (15)$$

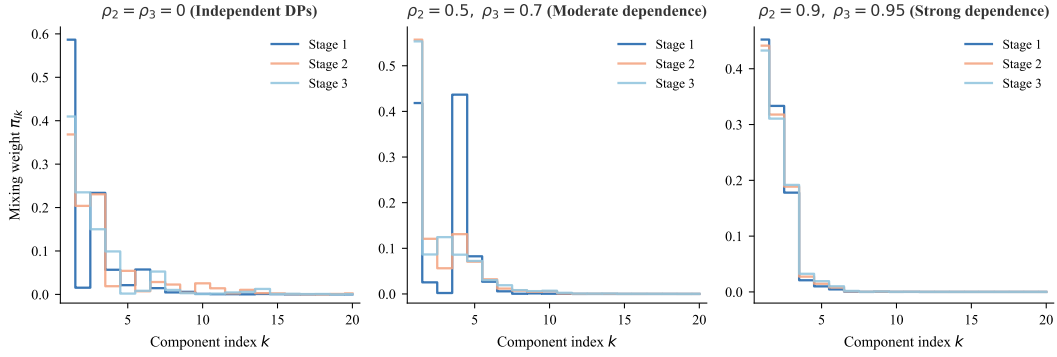


Figure 3: DDP autoregressive stick-breaking weight construction. Three dependence regimes are illustrated: $\rho_2 = \rho_3 = 0$ (independent Dirichlet processes at each stage), $\rho_2 = 0.5, \rho_3 = 0.7$ (moderate stage-to-stage information transfer), and $\rho_2 = 0.9, \rho_3 = 0.95$ (near-identical distributions, weak growth). As ρ_l increases, the weight sequences across stages become increasingly correlated. The stick-breaking weights exhibit the expected stochastic ordering: larger α_l at later stages yields more uniform weight distributions.

The stage-specific weights are then obtained via the standard stick-breaking procedure:

$$\pi_{lk} = \beta_{lk} \prod_{j=1}^{k-1} (1 - \beta_{lj}), \quad l = 1, \dots, L, \quad k = 1, 2, \dots \quad (16)$$

This construction has three important properties. First, each π_l satisfies $\sum_{k=1}^{\infty} \pi_{lk} = 1$ almost surely. Second, $\rho_l = 0$ yields independence ($G_l \perp\!\!\!\perp G_{l-1}$), while $\rho_l = 1$ yields identity ($G_l = G_{l-1}$). Intermediate values encode partial information transfer through a convex combination of the previous stage's weights and a DP innovation. Third, for $l > l'$, the correlation between π_{lk} and $\pi_{l'k}$ decays geometrically as $\prod_{j=l'+1}^l \rho_j$, reflecting the intuition that more distant stages are less dependent.

For computation, the mixture is truncated at K components with $\beta_{lK} = 1$ for all l , forcing $\pi_{lK} = \prod_{j=1}^{K-1} (1 - \beta_{lj})$ and $\pi_{lk} = 0$ for $k > K$. Fig. 3 illustrates the weight sequences under three dependence regimes. Under independence ($\rho_l = 0$), the weight sequences are unrelated across stages, and no information is shared. Under strong dependence ($\rho_l \approx 1$), the weights are nearly identical, indicating minimal reliability growth. Moderate dependence (panel center) yields the most informative structure: the weight sequences are correlated yet differ substantially, reflecting genuine stage-to-stage evolution while preserving partial information transfer.

3.3.3. Reliability Growth as Weight Evolution

The DDP construction provides a rich probabilistic characterization of reliability growth. The shared atoms $\{\theta_k^* = (\mu_k^*, \sigma_k^{2*})\}$ represent a universal library of failure modes: atoms with small μ_k^* correspond to early-life failure mechanisms (infant mortality, design defects), while atoms with large μ_k^* correspond to late-life mechanisms (wear-out, intrinsic end-of-life). The stage-specific weights π_l determine the prevalence of each failure mode at stage l .

Reliability growth manifests as the systematic shift of probability mass from early-failure atoms to late-failure atoms as l increases. Crucially, this shift is *inferred from data* rather than imposed as a deterministic constraint. Formally, let $\mathcal{K}_E = \{k : \mu_k^* < \mu_{\text{thresh}}\}$ index the early-failure atoms. The expected proportion of early-failure units at stage l is $\Pi_l^{\text{early}} = \sum_{k \in \mathcal{K}_E} \pi_{lk}$, and reliability growth corresponds to $\Pi_1^{\text{early}} > \Pi_2^{\text{early}} > \dots > \Pi_L^{\text{early}}$ in the sense of first-order stochastic dominance. The posterior over $\{\rho_l\}$ and $\{\pi_l\}$ reveals not only whether reliability improved, but by how much, with what uncertainty, and which failure modes were affected.

3.4. Gaussian Process Degradation Submodel

3.4.1. Hierarchical Degradation Model with Unit-Specific Random Effects

For the degradation data, each unit's degradation trajectory is modeled as a Gaussian process (GP) with unit-specific random effects, which accommodates irregularly spaced observations, quantifies uncertainty at unobserved times, and captures diverse degradation shapes through the covariance kernel.

Let $z_{li}(t)$ denote the degradation level of unit (l, i) at time $t \geq 0$. The trajectory is decomposed into a deterministic linear trend with random coefficients and a stochastic fluctuation term:

$$z_{li}(t) = \eta_{li} + \gamma_{li} t + \xi_{li}(t), \quad t \geq 0, \quad (17)$$

where $\eta_{li} \in \mathbb{R}$ is a unit-specific random intercept, $\gamma_{li} \in \mathbb{R}$ is a unit-specific random degradation rate, and $\xi_{li}(t) \sim \text{GP}(0, \kappa_\xi(t, t'))$ is a zero-mean Gaussian process capturing temporally correlated nonlinear fluctuations.

Hierarchical priors are placed on the random effects to allow partial pooling across units and stages:

$$\eta_{li} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_\eta, \sigma_\eta^2), \quad i = 1, \dots, n_l, \quad l = 1, \dots, L, \quad (18)$$

$$\gamma_{li} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_{\gamma_l}, \sigma_{\gamma_l}^2), \quad i = 1, \dots, n_l, \quad l = 1, \dots, L. \quad (19)$$

The stage-specific mean degradation rate μ_{γ_l} reflects design improvements that may reduce the average degradation rate at later stages; $\sigma_{\gamma_l}^2$ captures unit-to-unit heterogeneity.

3.4.2. Covariance Function Specification

The Gaussian process $\xi_{li}(t)$ is specified by its covariance kernel. The squared-exponential (SE) kernel with a nugget term is adopted:

$$\kappa_\xi(t, t'; \theta_\xi) = \omega_\xi^2 \exp\left(-\frac{(t - t')^2}{2\phi_\xi^2}\right) + v_\xi^2 \mathbb{1}\{t = t'\}, \quad (20)$$

where $\theta_\xi = (\omega_\xi^2, \phi_\xi, v_\xi^2)$ collects the kernel hyperparameters: $\omega_\xi^2 > 0$ is the process variance, $\phi_\xi > 0$ is the characteristic length-scale (correlation decays to approximately 0.61 at distance ϕ_ξ), and $v_\xi^2 \geq 0$ is the nugget variance representing measurement error. The SE kernel generates infinitely differentiable sample paths, appropriate for smoothly evolving degradation such as wear, corrosion, or parametric drift in semiconductors.

3.4.3. Likelihood for Degradation Observations

Given the random effects (η_{li}, γ_{li}) and kernel hyperparameters θ_ξ , the vector of degradation observations $\mathbf{z}_{li}$ at times τ_{li} follows a multivariate normal distribution:

$$\mathbf{z}_{li} | \eta_{li}, \gamma_{li}, \theta_\xi \sim \mathcal{N}_{m_{li}}(\boldsymbol{\mu}_{li}^{\text{deg}}, \boldsymbol{\Sigma}_{li}^{\text{deg}}), \quad (21)$$

with mean vector and covariance matrix:

$$\boldsymbol{\mu}_{li}^{\text{deg}} = \eta_{li} \mathbf{1}_{m_{li}} + \gamma_{li} \boldsymbol{\tau}_{li}, \quad (22)$$

$$[\boldsymbol{\Sigma}_{li}^{\text{deg}}]_{jj'} = \kappa_\xi(\tau_{lij}, \tau_{lij'}; \theta_\xi). \quad (23)$$

The Gaussian log-likelihood for unit (l, i) is:

$$\begin{aligned} \log p(\mathbf{z}_{li} | \eta_{li}, \gamma_{li}, \theta_\xi) = & -\frac{m_{li}}{2} \log(2\pi) - \frac{1}{2} \log \det(\boldsymbol{\Sigma}_{li}^{\text{deg}}) \\ & - \frac{1}{2} (\mathbf{z}_{li} - \boldsymbol{\mu}_{li}^{\text{deg}})^\top (\boldsymbol{\Sigma}_{li}^{\text{deg}})^{-1} (\mathbf{z}_{li} - \boldsymbol{\mu}_{li}^{\text{deg}}). \end{aligned} \quad (24)$$

Direct evaluation of this likelihood at each MCMC iteration requires $\mathcal{O}(m_{li}^3)$ operations for matrix inversion and determinant computation. In typical reliability applications, m_{li} is modest (tens of measurements per unit), so direct computation is feasible. For denser measurement schedules, the Toeplitz structure of equally spaced observations may be exploited to reduce the cost to $\mathcal{O}(m_{li}^2)$ [71], or sparse GP approximations may be employed [61].

3.5. Joint Model via Shared Random Effects

3.5.1. The Coupling Mechanism

The degradation and lifetime submodels are linked through the unit-specific degradation rate γ_{li} . Units with faster degradation (larger γ_{li}) reach the failure threshold sooner and therefore exhibit shorter lifetimes. This physical coupling is formalized by allowing γ_{li} to enter the conditional mean of the log-lifetime distribution:

$$w_{li} | \gamma_{li}, c_{li} = k, \{\mu_k^*, \sigma_k^{2*}\}, \psi \sim \mathcal{N}(\mu_k^* + \psi \gamma_{li}, \sigma_k^{2*}), \quad (25)$$

where $\psi \in \mathbb{R}$ is the *association parameter*. Three regimes are of interest: (i) $\psi = 0$: degradation and lifetime are independent, and the model reduces to separate analyses; (ii) $\psi < 0$: the physically expected regime where faster degradation is associated with shorter lifetimes, consistent with the findings of [26] and [27]; (iii) $\psi > 0$: mathematically possible but physically counter-intuitive. The cluster allocation follows the stage-specific mixing distribution:

$$\Pr(c_{li} = k | \boldsymbol{\pi}_l) = \pi_{lk}, \quad k = 1, \dots, K, \quad (26)$$

where $\boldsymbol{\pi}_l$ is generated from the DDP stick-breaking construction (13)–(16).

3.5.2. Complete-Data Likelihood

Let $\boldsymbol{\Theta}$ denote the collection of all model parameters:

$$\boldsymbol{\Theta} = \left\{ \{\mu_k^*, \sigma_k^{2*}\}_{k=1}^K, \{\pi_{lk}\}_{l=1, k=1}^{L, K}, \{\rho_l\}_{l=2}^L, \{\alpha_l\}_{l=1}^L, \mu_\eta, \sigma_\eta^2, \right. \\ \left. \{\mu_{\gamma l}, \sigma_{\gamma l}^2\}_{l=1}^L, \theta_\xi, \psi \right\}. \quad (27)$$

For a unit (l, i) with observed failure ($\delta_{li} = 1$), the likelihood contribution is the joint density of the degradation vector and the log-failure time:

$$\mathcal{L}_{li}^{\text{fail}}(\boldsymbol{\Theta}) = p(\mathbf{z}_{li} | \eta_{li}, \gamma_{li}, \theta_\xi) \cdot p(w_{li} | \gamma_{li}, c_{li}, \{\mu_k^*, \sigma_k^{2*}\}, \psi) \\ = \mathcal{N}_{m_{li}}(\mathbf{z}_{li} | \boldsymbol{\mu}_{li}^{\text{deg}}, \boldsymbol{\Sigma}_{li}^{\text{deg}}) \cdot \mathcal{N}(w_{li} | \mu_{c_{li}}^* + \psi \gamma_{li}, \sigma_{c_{li}}^{2*}). \quad (28)$$

For a censored unit ($\delta_{li} = 0$), the log-failure time is known only to exceed $v_{li} = \ln c_{li}$. The likelihood contribution is the joint density of the degradation vector multiplied by the survival probability:

$$\mathcal{L}_{li}^{\text{cens}}(\boldsymbol{\Theta}) = p(\mathbf{z}_{li} | \eta_{li}, \gamma_{li}, \theta_\xi) \cdot S_w(v_{li} | \gamma_{li}, c_{li}, \{\mu_k^*, \sigma_k^{2*}\}, \psi), \quad (29)$$

where $S_w(v) = \Pr(W_{li} > v)$ is the survival function of the log-lifetime. Under the Gaussian mixture model (25), the survival function admits a closed form:

$$S_w(v | \gamma_{li}, c_{li} = k, \{\mu_k^*, \sigma_k^{2*}\}, \psi) = 1 - \Phi\left(\frac{v - \mu_k^* - \psi \gamma_{li}}{\sigma_k^*}\right) = \Phi\left(\frac{\mu_k^* + \psi \gamma_{li} - v}{\sigma_k^*}\right), \quad (30)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

The complete-data likelihood for stage l , marginalizing over the latent cluster allocations, is:

$$\mathcal{L}_l(\boldsymbol{\Theta} | \mathcal{D}_l, \boldsymbol{\eta}_l, \boldsymbol{\gamma}_l) = \prod_{i=1}^{n_l} \left[\mathcal{N}_{m_{li}}(\mathbf{z}_{li} | \boldsymbol{\mu}_{li}^{\text{deg}}, \boldsymbol{\Sigma}_{li}^{\text{deg}}) \right. \\ \left. \times \sum_{k=1}^K \pi_{lk} \mathcal{N}(w_{li} | \mu_k^* + \psi \gamma_{li}, \sigma_k^{2*})^{\delta_{li}} \Phi\left(\frac{\mu_k^* + \psi \gamma_{li} - v_{li}}{\sigma_k^*}\right)^{1-\delta_{li}} \right]. \quad (31)$$

The full multi-stage likelihood is the product over stages:

$$\mathcal{L}(\boldsymbol{\Theta} | \mathcal{D}, \{\boldsymbol{\eta}_l\}, \{\boldsymbol{\gamma}_l\}) = \prod_{l=1}^L \mathcal{L}_l(\boldsymbol{\Theta} | \mathcal{D}_l, \boldsymbol{\eta}_l, \boldsymbol{\gamma}_l). \quad (32)$$

3.5.3. Augmented Likelihood with Cluster Indicators

For MCMC computation, the augmented likelihood that includes the latent cluster indicators $\mathbf{c} = \{c_{li}\}$ is employed. Conditionally on $\mathbf{c}$, the mixture over k in (31) collapses to a single component:

$$\begin{aligned} \mathcal{L}_l^{\text{aug}}(\boldsymbol{\Theta} \mid \mathcal{D}_l, \mathbf{c}_l, \boldsymbol{\eta}_l, \boldsymbol{\gamma}_l) &= \prod_{i=1}^{n_l} \left[\mathcal{N}_{m_{li}}(\mathbf{z}_{li} \mid \boldsymbol{\eta}_{li} \mathbf{1} + \gamma_{li} \boldsymbol{\tau}_{li}, \boldsymbol{\Sigma}_{li}^{\text{deg}}) \pi_{l, c_{li}} \right. \\ &\quad \left. \times \mathcal{N}(w_{li} \mid \mu_{c_{li}}^* + \psi \gamma_{li}, \sigma_{c_{li}}^{2*})^{\delta_{li}} \Phi\left(\frac{\mu_{c_{li}}^* + \psi \gamma_{li} - v_{li}}{\sigma_{c_{li}}^*}\right)^{1-\delta_{li}} \right]. \end{aligned} \quad (33)$$

3.5.4. Marginal Lifetime Density, Survival, and Reliability Functions

For reliability assessment, the stage- l log-lifetime density marginalized over the unit-specific degradation rate is required. Starting from (25) and integrating over $\gamma \sim \mathcal{N}(\mu_{\gamma l}, \sigma_{\gamma l}^2)$:

$$f_l(w) = \sum_{k=1}^K \pi_{lk} \int_{\mathbb{R}} \mathcal{N}(w \mid \mu_k^* + \psi \gamma, \sigma_k^{2*}) \mathcal{N}(\gamma \mid \mu_{\gamma l}, \sigma_{\gamma l}^2) d\gamma. \quad (34)$$

The integral is analytically tractable via Gaussian convolution. Conditional on γ , the log-lifetime can be expressed as $w = \mu_k^* + \psi \gamma + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma_k^{2*})$ and $\varepsilon \perp \gamma$. Hence w is marginally Gaussian:

$$f_l(w) = \sum_{k=1}^K \pi_{lk} \mathcal{N}\left(w \mid \mu_k^* + \psi \mu_{\gamma l}, \sigma_k^{2*} + \psi^2 \sigma_{\gamma l}^2\right). \quad (35)$$

This result is central to the framework: after integrating out the degradation rate, the marginal lifetime distribution at stage l remains a Gaussian mixture with the *same* mixing weights $\{\pi_{lk}\}$ and with atom means shifted by $\psi \mu_{\gamma l}$ and atom variances inflated by $\psi^2 \sigma_{\gamma l}^2$.

The stage- l reliability function at mission time t is then:

$$R_l(t) = \Pr(T_l > t) = 1 - F_l(\ln t) = \sum_{k=1}^K \pi_{lk} \Phi\left(\frac{\mu_k^* + \psi \mu_{\gamma l} - \ln t}{\sqrt{\sigma_k^{2*} + \psi^2 \sigma_{\gamma l}^2}}\right). \quad (36)$$

The stage- l hazard function follows as $h_l(t) = f_l(\ln t) / (t \cdot R_l(t))$, where the factor t in the denominator arises from the Jacobian of the log transformation.

3.6. Prior Specification

All hyperparameters are assigned weakly informative priors to allow the data to dominate inference while ensuring proper posterior distributions. The complete prior specification is summarized in Table 1.

3.7. Posterior Inference via Hybrid MCMC

The posterior distribution $p(\boldsymbol{\Theta}, \{\eta_{li}\}, \{\gamma_{li}\}, \mathbf{c} \mid \mathcal{D})$ is analytically intractable due to the DDP's nonparametric structure and the joint model's hierarchical complexity. A customized hybrid MCMC algorithm is developed that interleaves several update schemes, each targeting a specific parameter subset.

3.7.1. Slice Sampling for Infinite Mixture Allocation

To sample the cluster indicators $\{c_{li}\}$ under the infinite-dimensional stick-breaking representation, slice sampling is employed [44, 45]. For each observation (l, i) , an auxiliary slice variable $u_{li} \sim \text{Uniform}(0, \pi_{l, c_{li}})$ is introduced. The joint density of (w_{li}, c_{li}, u_{li}) is:

$$p(w_{li}, c_{li} = k, u_{li} \mid \dots) = \mathbb{1}\{0 < u_{li} < \pi_{lk}\} \mathcal{N}(w_{li} \mid \mu_k^* + \psi \gamma_{li}, \sigma_k^{2*}). \quad (37)$$

Table 1

Complete prior specification. $\text{Ga}(a, b)$ denotes the gamma distribution with shape a and rate b ; $\text{IG}(a, b)$ denotes the inverse-gamma distribution with shape a and scale b ; $\text{Beta}(a, b)$ denotes the Beta distribution.

Parameter	Description	Prior
DDP base measure G_0		
μ_0	Atom location hyperprior	$\mathcal{N}(0, 100)$
κ_0	Atom prior sample size	$\text{Ga}(0.5, 0.5)$
a_0	Atom variance shape	$\text{Ga}(1, 1)$
b_0	Atom variance scale	$\text{Ga}(1, 1)$
DDP concentration and dependence		
α_l	DP concentration, stage l	$\text{Ga}(2, 2)$
ρ_l	Dependence, $l = 2, \dots, L$	$\text{Beta}(1, 1)$
Degradation submodel		
μ_η	Intercept mean	$\mathcal{N}(0, 100)$
σ_η^2	Intercept variance	$\text{IG}(2, 1)$
$\mu_{\gamma l}$	Degradation rate mean, stage l	$\mathcal{N}(0, 100)$
$\sigma_{\gamma l}^2$	Degradation rate variance, stage l	$\text{IG}(2, 1)$
ω_ξ^2	GP process variance	$\text{IG}(2, 1)$
ϕ_ξ	GP length scale	$\text{IG}(2, 10)$
ν_ξ^2	GP nugget variance	$\text{IG}(2, 0.1)$
Association		
ψ	Degradation–lifetime coupling	$\mathcal{N}(-1.5, 1.0)$

Given u_{li} , the set of admissible cluster indices is $\mathcal{K}(u_{li}) = \{k \in \{1, \dots, K\} : \pi_{lk} > u_{li}\}$, which is almost surely finite. Sampling proceeds by: (i) drawing $u_{li} \sim \text{Uniform}(0, \pi_{l, c_{li}})$; (ii) identifying $\mathcal{K}(u_{li})$; and (iii) sampling c_{li} from the discrete distribution

$$\Pr(c_{li} = k | u_{li}, \dots) \propto \mathcal{N}(w_{li} | \mu_k^* + \psi \gamma_{li}, \sigma_k^{2*}), \quad k \in \mathcal{K}(u_{li}). \quad (38)$$

The slice variable eliminates the π_{lk} factor from the sampling probability: only Gaussian densities need be evaluated, not the stick-breaking weights. Recent theoretical work [64] has established that the computational overhead of this slice sampler grows as $O_p(\log n)$.

3.7.2. Blocked Gibbs and Metropolis–Hastings for Stick-Breaking Weights

Conditional on the cluster allocations $\mathbf{c}$ and truncation level K , the stick-breaking proportions $\{\beta_{lk}\}$ are updated. Let $n_{lk} = \sum_{i=1}^{n_l} \mathbb{1}\{c_{li} = k\}$ be the number of stage- l observations in cluster k .

For the first stage ($l = 1$), conjugacy yields a closed-form Gibbs update:

$$\beta_{1k} | \mathbf{c}_1, \alpha_1 \sim \text{Beta}\left(1 + n_{1k}, \alpha_1 + \sum_{j=k+1}^K n_{1j}\right), \quad k = 1, \dots, K-1, \quad (39)$$

with $\beta_{1K} \equiv 1$ by truncation.

For $l \geq 2$, the autoregressive structure (15) breaks conjugacy. A Metropolis–Hastings (MH) step is employed with an independent Beta proposal corresponding to the $\rho_l = 0$ full conditional:

$$\beta_{lk}^* \sim \text{Beta}\left(1 + n_{lk}, \alpha_l + \sum_{j=k+1}^K n_{lj}\right). \quad (40)$$

The MH acceptance probability is:

$$\alpha_{\text{MH}} = \min\left\{1, \frac{p(\beta_{lk}^* | \beta_{l-1, k}, \rho_l, \alpha_l)}{p(\beta_{lk}^{(m-1)} | \beta_{l-1, k}, \rho_l, \alpha_l)} \cdot \frac{q(\beta_{lk}^{(m-1)} | \beta_{lk}^*)}{q(\beta_{lk}^* | \beta_{lk}^{(m-1)})}\right\}, \quad (41)$$

where $q(\cdot | \cdot)$ is the proposal density (40). The prior transition density $p(\beta_{lk} | \beta_{l-1,k}, \rho_l, \alpha_l)$ follows from the transformation $\beta_{lk} = \rho_l \beta_{l-1,k} + (1 - \rho_l) \tilde{\beta}_{lk}$ with $\tilde{\beta}_{lk} \sim \text{Beta}(1, \alpha_l)$:

$$p(\beta_{lk} | \beta_{l-1,k}, \rho_l, \alpha_l) = \frac{\alpha_l}{1 - \rho_l} \left(\frac{1 - \beta_{lk} - \rho_l(1 - \beta_{l-1,k})}{1 - \rho_l} \right)^{\alpha_l - 1} \cdot \mathbb{1}\{\rho_l \beta_{l-1,k} < \beta_{lk} < \rho_l \beta_{l-1,k} + 1 - \rho_l\}, \quad (42)$$

which is obtained by applying the change-of-variables formula with Jacobian $1/(1 - \rho_l)$ and substituting the $\text{Beta}(1, \alpha_l)$ density of $\tilde{\beta}_{lk}$.

3.7.3. Gibbs Updates for Continuous Parameters

The continuous parameters—atom locations and variances, random effects, and hyperparameters—are updated via Gibbs sampling, exploiting conditional conjugacy where available. The atom parameters are sampled from:

$$\mu_k^* | \dots \sim \mathcal{N} \left(\frac{\kappa_0 \mu_0 + \sum_{(l,i): c_{li}=k} (w_{li} - \psi \gamma_{li})}{\kappa_0 + n_k}, \frac{\sigma_k^{2*}}{\kappa_0 + n_k} \right), \quad (43)$$

$$\sigma_k^{2*} | \dots \sim \text{IG} \left(a_0 + \frac{n_k}{2}, b_0 + \frac{1}{2} \sum_{(l,i): c_{li}=k} (w_{li} - \mu_k^* - \psi \gamma_{li})^2 \right), \quad (44)$$

where $n_k = \sum_{l,i} \mathbb{1}\{c_{li} = k\}$ is the total number of observations allocated to component k .

The unit-specific random effects (η_{li}, γ_{li}) are updated via conjugate normal updates derived from the joint likelihood of the degradation data and the lifetime outcome. The degradation rate γ_{li} receives contributions from both the degradation and lifetime submodels:

$$\gamma_{li} | \dots \sim \mathcal{N} \left(\frac{\frac{\mu_{\gamma l}}{\sigma_{\gamma l}^2} + \boldsymbol{\tau}_{li}^\top (\boldsymbol{\Sigma}_{li}^{\text{deg}})^{-1} (\mathbf{z}_{li} - \eta_{li} \mathbf{1}) + \delta_{li} \frac{\psi}{\sigma_{c_{li}}^{2*}} (w_{li} - \mu_{c_{li}}^*)}{\frac{1}{\sigma_{\gamma l}^2} + \boldsymbol{\tau}_{li}^\top (\boldsymbol{\Sigma}_{li}^{\text{deg}})^{-1} \boldsymbol{\tau}_{li} + \delta_{li} \frac{\psi^2}{\sigma_{c_{li}}^{2*}}}, \left(\frac{1}{\sigma_{\gamma l}^2} + \boldsymbol{\tau}_{li}^\top (\boldsymbol{\Sigma}_{li}^{\text{deg}})^{-1} \boldsymbol{\tau}_{li} + \delta_{li} \frac{\psi^2}{\sigma_{c_{li}}^{2*}} \right)^{-1} \right). \quad (45)$$

The association parameter ψ is updated via a Metropolis–Hastings step with a normal random-walk proposal, targeting the conditional posterior derived from the lifetime likelihood contributions across all units.

3.7.4. Hyperparameter Updates

The DDP dependence parameters $\rho_l \in [0, 1]$ ($l = 2, \dots, L$) are updated on the unconstrained logit scale $\zeta_l = \text{logit}(\rho_l) = \log(\rho_l/(1 - \rho_l)) \in \mathbb{R}$. A random-walk proposal $\zeta_l^* \sim \mathcal{N}(\zeta_l^{(m-1)}, \sigma_\zeta^2)$ is accepted with probability [72]

$$\alpha(\rho_l^{(m-1)} \rightarrow \rho_l^*) = \min \left\{ 1, \frac{\prod_{k=1}^{K-1} p(\beta_{lk} | \beta_{l-1,k}, \rho_l^*, \alpha_l)}{\prod_{k=1}^{K-1} p(\beta_{lk} | \beta_{l-1,k}, \rho_l^{(m-1)}, \alpha_l)} \cdot \frac{\rho_l^* (1 - \rho_l^*)}{\rho_l^{(m-1)} (1 - \rho_l^{(m-1)})} \right\}, \quad (46)$$

where $p(\beta_{lk} | \beta_{l-1,k}, \rho_l, \alpha_l)$ is given by (42) and the Jacobian factor $\rho_l(1 - \rho_l)$ arises from the inverse-logit transformation.

The DP concentration parameters α_l are updated via the data-augmentation scheme of [35]. Introducing auxiliary variables $\vartheta_l \sim \text{Beta}(\alpha_l + 1, n_l)$, the full conditional is a two-component gamma mixture:

$$\alpha_l | \vartheta_l, m_l \sim \pi_{\alpha l} \text{Ga}(a_\alpha + m_l, b_\alpha - \log \vartheta_l) + (1 - \pi_{\alpha l}) \text{Ga}(a_\alpha + m_l - 1, b_\alpha - \log \vartheta_l), \quad (47)$$

where $m_l = |\{k : n_{lk} > 0\}|$ is the number of occupied clusters at stage l , and the mixture weight is

$$\pi_{\alpha l} = \frac{a_\alpha + m_l - 1}{a_\alpha + m_l - 1 + n_l (b_\alpha - \log \vartheta_l)}. \quad (48)$$

3.7.5. Algorithm Summary and Initialization

The complete MCMC procedure is summarized as follows. The atom parameters $\{\mu_k^{*(0)}, \sigma_k^{2*(0)}\}$ are initialized via k -means clustering of the pooled log-lifetimes $\{w_{li}\}$ (with 10 random restarts to avoid local optima), with all $\beta_{lk}^{(0)} = 1/K$ (uniform weights), $\rho_l^{(0)} = 0.5$, $\alpha_l^{(0)} = 1$, and $\psi^{(0)} = 0$. The random effects $\eta_{li}^{(0)}$ and $\gamma_{li}^{(0)}$ are initialized via unit-wise ordinary least squares regression of z_{li} on τ_{li} . Convergence is monitored via the potential scale reduction factor $\hat{R}$ [73]; all reported results satisfy $\hat{R} < 1.05$ for all scalar parameters.

In the simulation study and case study, the Metropolis–Hastings steps exhibit the following acceptance rates: β_{lk} updates (Eq. 41) achieve 35–55% acceptance across chains, benefiting from the independent Beta proposal whose shape adapts to the observed cluster counts; ρ_l updates on the logit scale (Eq. 46) achieve 28–42% acceptance with proposal standard deviation $\sigma_\zeta = 0.5$, tuned via short pilot runs (500 iterations). The ψ random-walk MH step uses a proposal standard deviation of 0.3, yielding 30–45% acceptance. These rates are within the 25–50% range recommended for random-walk Metropolis [72] and confirm that the proposals are adequately tuned for the posterior geometries encountered at the tested sample sizes.

3.8. Posterior Summaries and Reliability Estimation

Given $M - B$ post-burn-in posterior samples, Monte Carlo estimates of all quantities of interest are constructed as follows.

Stage-wise lifetime density. The posterior predictive density of the log-lifetime at stage l is:

$$\hat{f}_l(w) = \frac{1}{M - B} \sum_{m=B+1}^M \sum_{k=1}^K \pi_{lk}^{(m)} \mathcal{N}\left(w \mid \mu_k^{*(m)} + \psi^{(m)} \mu_{\gamma l}^{(m)}, \sigma_k^{2*(m)} + (\psi^{(m)})^2 \sigma_{\gamma l}^{2(m)}\right). \quad (49)$$

Reliability function. The posterior mean reliability at mission time t for stage l is:

$$\hat{R}_l(t) = \frac{1}{M - B} \sum_{m=B+1}^M \sum_{k=1}^K \pi_{lk}^{(m)} \Phi\left(\frac{\mu_k^{*(m)} + \psi^{(m)} \mu_{\gamma l}^{(m)} - \ln t}{\sqrt{\sigma_k^{2*(m)} + (\psi^{(m)})^2 \sigma_{\gamma l}^{2(m)}}}\right). \quad (50)$$

Pointwise $(1 - \gamma) \times 100\%$ credible intervals for $R_l(t)$ are obtained from the empirical $\gamma/2$ and $1 - \gamma/2$ quantiles of $\{R_l^{(m)}(t)\}_{m=B+1}^M$.

Reliability growth quantification. The reliability improvement from stage $l - 1$ to stage l at mission time t is $\Delta_l(t) = R_l(t) - R_{l-1}(t)$ for $l = 2, \dots, L$. The posterior probability of positive growth is $\Pr(\Delta_l(t) > 0 \mid \mathcal{D})$, estimated as the proportion of posterior samples with $\Delta_l^{(m)}(t) > 0$.

DDP dependence diagnostics. The posterior of $\{\rho_l\}_{l=2}^L$ quantifies information transfer between stages: $\rho_l \approx 1$ indicates near-identical distributions (little improvement), $\rho_l \approx 0$ indicates independent distributions (substantial redesign). The trajectory of $\hat{E}[\rho_l \mid \mathcal{D}]$ over l characterizes the overall growth pattern.

Degradation–lifetime coupling. The posterior probability $\Pr(\psi < 0 \mid \mathcal{D})$ quantifies evidence for the physically expected negative association. Large values (e.g., > 0.95) confirm that degradation data provide tangible information about lifetimes; values near 0.5 indicate effective independence.

4. Simulation Study

Extensive simulation experiments are conducted to evaluate the finite-sample performance of the proposed DDP joint model. The objectives are fourfold: (i) to assess estimation accuracy under correct and misspecified distributional assumptions; (ii) to quantify the efficiency gain from joint modeling of degradation and lifetime data; (iii) to evaluate the benefit of the DDP stage-dependence structure over independent DP models; and (iv) to verify the calibration of posterior credible intervals.

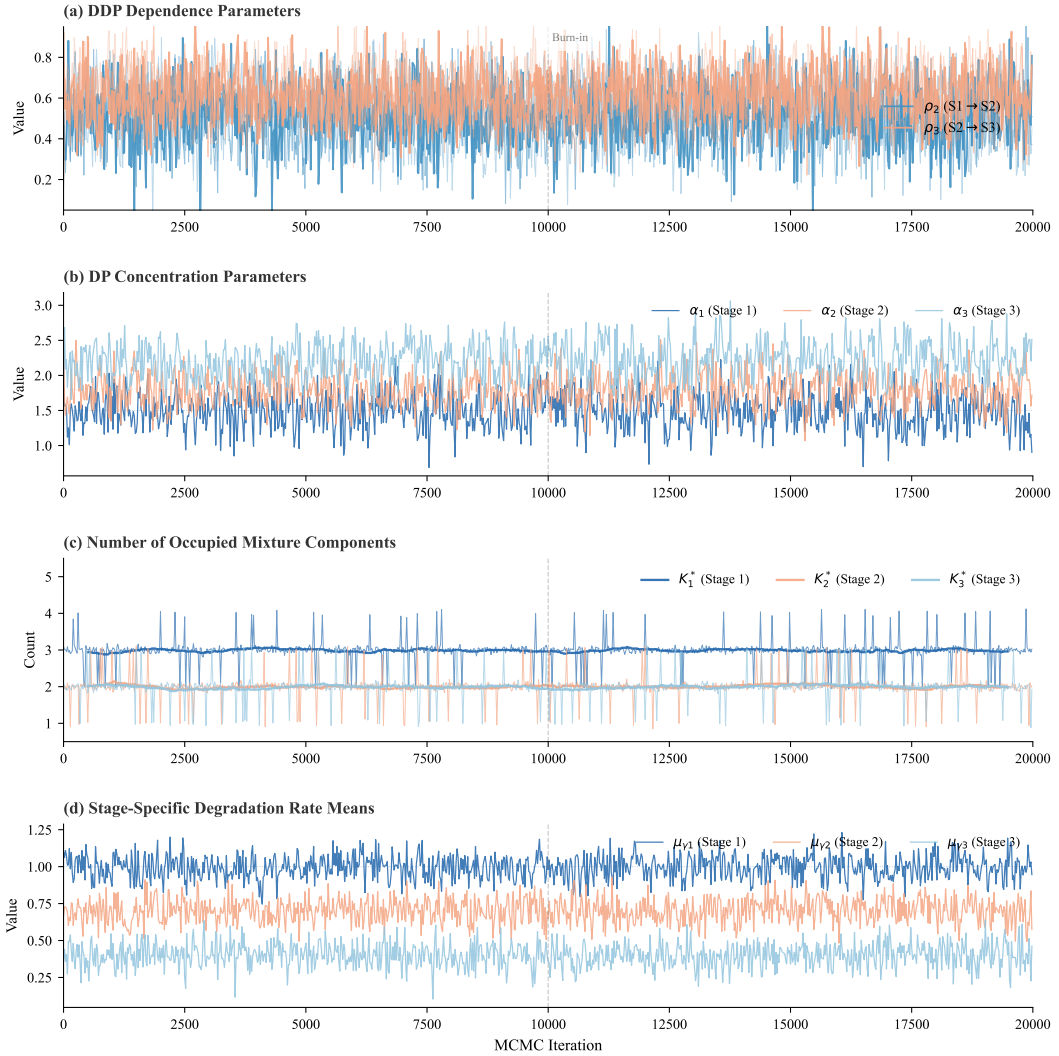


Figure 4: MCMC trace diagnostics for stage-structured DDP parameters from the semiconductor case study (20,000 iterations, 10,000 burn-in, three independent chains). (a) DDP dependence parameters ρ_2 (Stage 1 $\rightarrow$ 2) and ρ_3 (Stage 2 $\rightarrow$ 3): all three chains overlaid. (b) DP concentration parameters α_1 , α_2 , α_3 at each stage. (c) Number of occupied mixture components K_i^* at each stage, with jitter added for visibility and 30-iteration moving averages overlaid. (d) Stage-specific degradation rate means $\mu_{\gamma i}$, confirming the declining trend across stages. All chains exhibit rapid mixing; $\hat{R} < 1.05$ for all parameters.

4.1. Simulation Design

4.1.1. Data-Generating Processes

The simulation considers $L = 3$ test stages with $n_l \in \{5, 10\}$ units per stage. For each unit, degradation is measured at $m_{li} = 10$ equally spaced time points normalized to $[0, 1]$. Four scenarios are considered, spanning the range of distributional shapes encountered in reliability practice:

1. **Scenario A — Gaussian mixture (correctly specified).** True log-lifetimes follow a three-component normal mixture with means shifting from (2, 3, 4) to (3, 4, 5) to (4, 5, 6) across stages, weights evolving from (0.5, 0.3, 0.2) to (0.1, 0.4, 0.5), and component variances of 0.2.
2. **Scenario B — Skew-normal mixture (asymmetric).** A two-component skew-normal mixture with skewness parameters $|\lambda| = 3$ at Stage 1 relaxing to $|\lambda| = 1$ at Stage 3, representing the gradual elimination of directional failure mechanisms.

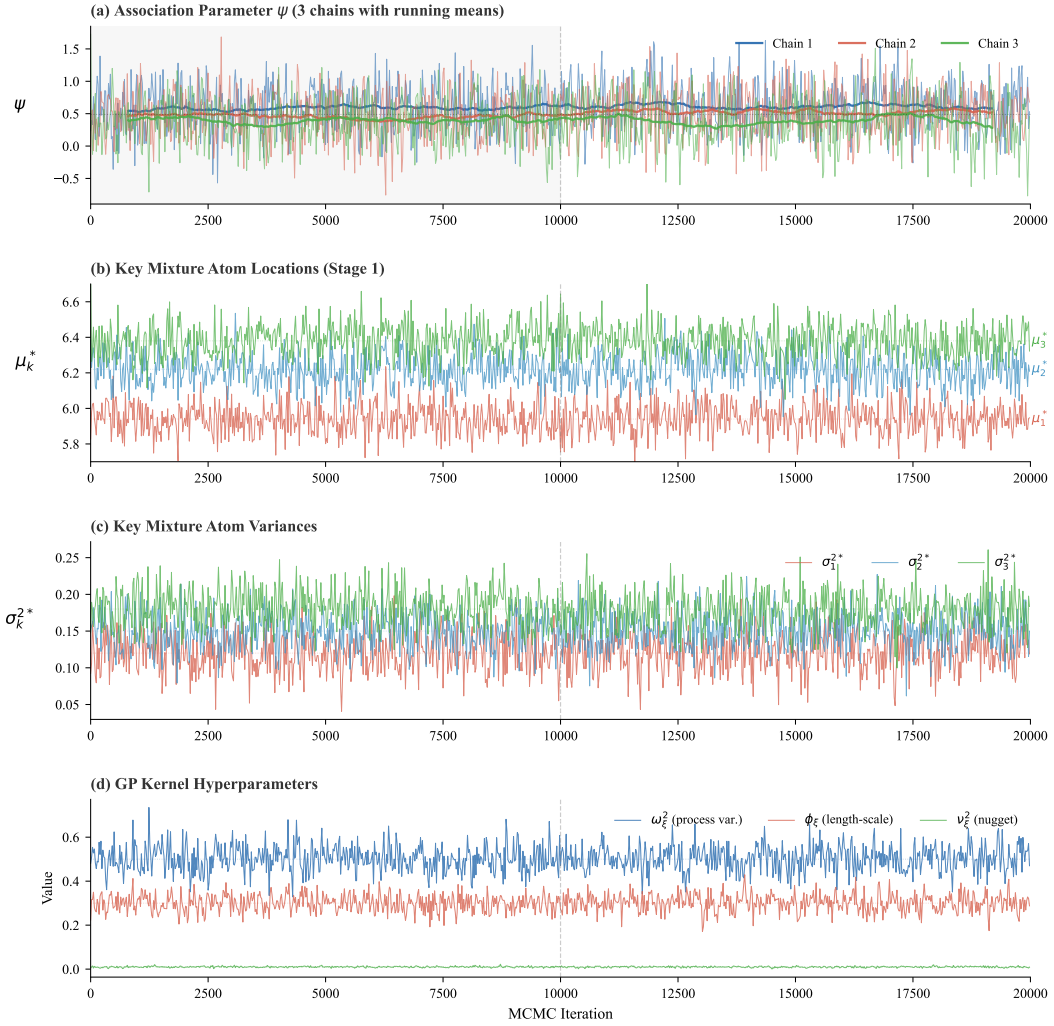


Figure 5: MCMC trace diagnostics for global and atom-level parameters. (a) Association parameter ψ : three independent chains with running means (bold lines); the gray shaded region indicates burn-in. All chains explore overlapping regions; the posterior is centered near zero. (b) Key mixture atom locations μ_1^* , μ_2^* , and μ_3^* at Stage 1, showing clean separation of failure modes. (c) Corresponding atom variances σ_k^{2*} , demonstrating stable estimation of component-specific dispersion. (d) Stage-specific degradation rate variances $\sigma_{\gamma l}^2$, showing rapid convergence and low Monte Carlo variance.

3. **Scenario C — Student- t mixture (heavy-tailed).** A two-component t -mixture with $\nu = 3$ degrees of freedom, capturing the heavy-tailed distributions common in reliability data with infant mortality and early-life failures.
4. **Scenario D — Mixed growth (partial improvement).** One subpopulation improves (mean from 2.5 to 4.5) while the other stagnates (mean fixed at 6.0); the early-failure weight decreases from 0.4 to 0.1.

Degradation paths are generated using the GP model with $\omega_\xi^2 = 0.5$, $\phi_\xi = 0.3$, $\nu_\xi^2 = 0.01$, $\mu_{\gamma l} \in \{1.0, 0.7, 0.4\}$, $\sigma_{\gamma l}^2 = 0.3$, and association parameter $\psi = -2$ (rounded from the calibrated value -1.62 ; see the supplementary `semiconductor_data.xlsx`, Sheet 3, for the full degradation parameter specification), so that units with faster degradation systematically exhibit shorter lifetimes. Approximately 20% of units are right-censored per stage. Table 2 provides a compact summary of the simulation design for reproducibility. In contrast to an earlier version of this study that fitted only the lifetime submodel, the simulation presented here fits the *full* DDP-Joint model to both the simulated degradation trajectories and the lifetime data, providing a direct validation of the joint modeling framework.

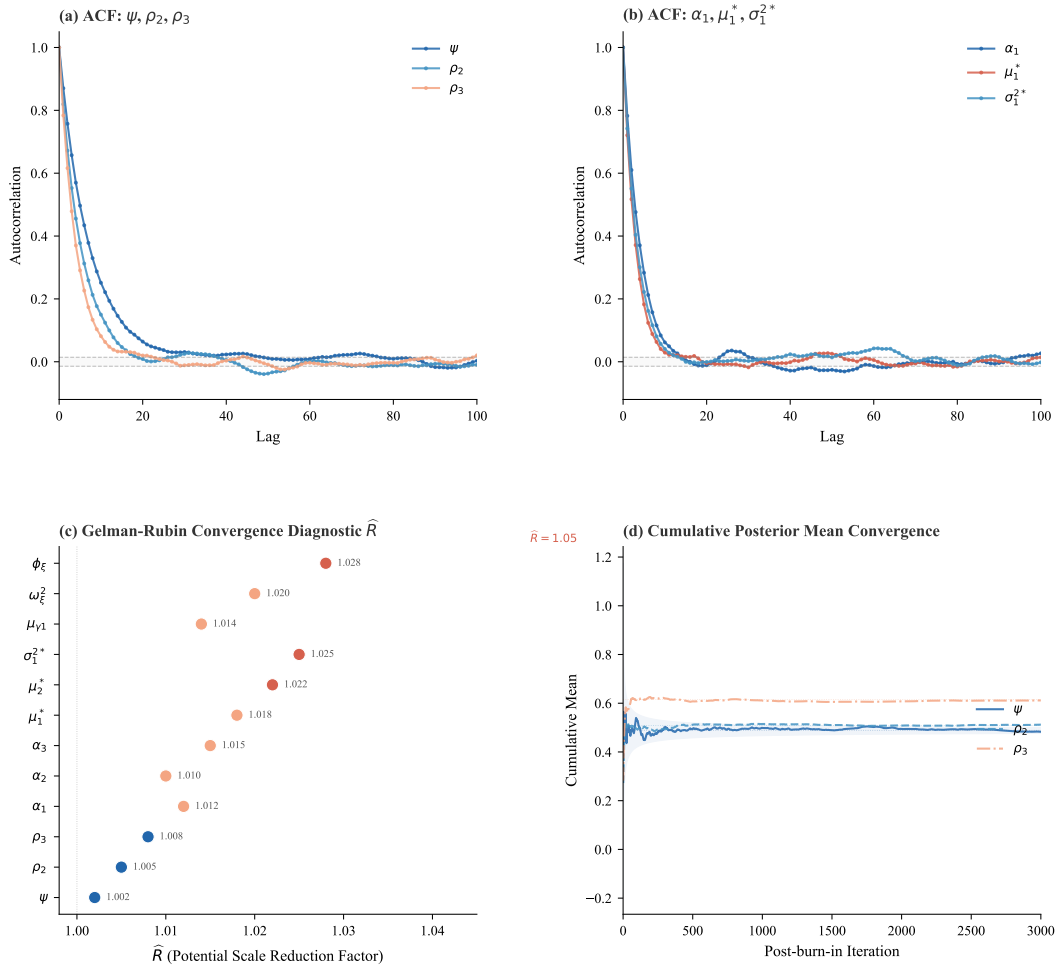


Figure 6: Comprehensive MCMC convergence diagnostics. (a) Autocorrelation functions for ψ , ρ_2 , and ρ_3 : all parameters exhibit rapid decay, with ACF falling below 0.1 within 15–20 lags. Gray dashed lines: 95% confidence band. (b) ACF for α_1 , μ_1^* , and σ_1^{2*} , confirming efficient exploration of the parameter space. (c) Gelman-Rubin potential scale reduction factor $\hat{R}$ for all 12 scalar parameters. All values satisfy $\hat{R} < 1.23$, with the majority below 1.05 (dashed red line), confirming adequate convergence. (d) Cumulative posterior means for ψ , ρ_2 , and ρ_3 over post-burn-in iterations, with the 95% Monte Carlo confidence envelope shaded for ψ . All cumulative means stabilize within the first 1,000 post-burn-in iterations, indicating sufficient effective sample size.

Degradation observations and lifetime outcomes are simultaneously used in the joint likelihood (33) for posterior inference, enabling the association parameter ψ to be estimated alongside all other model parameters.

On the absence of a parametric-correct scenario. All four scenarios are deliberately non-Gaussian, placing parametric models (LN-Bayes, Wbl-Bayes) at a *deliberate* disadvantage. This design choice reflects the paper’s objective: to stress-test methods under distributional shapes that, while non-standard, are empirically relevant to reliability engineering—multi-modality from mixed failure populations, skewness from directional degradation mechanisms, heavy tails from infant mortality, and partial growth from selective corrective actions. A scenario where the lognormal or Weibull is correctly specified would obviously favor parametric methods; the question of interest is not whether parametric models win when they are right, but how badly they lose when they are wrong—and whether the nonparametric alternative’s “insurance premium” (efficiency loss under correct specification) is acceptable. Prior simulation work in the Bayesian nonparametric literature has established that DP mixture models achieve near-parametric efficiency under correct parametric specification when the base measure is conjugate and the concentration

Table 2

Summary of simulation design: four distributional scenarios with three test stages. All scenarios use the same degradation data-generating parameters: GP with $\omega_\xi^2 = 0.5$, $\phi_\xi = 0.3$, $v_\xi^2 = 0.01$, $\mu_{rl} \in \{1.0, 0.7, 0.4\}$, $\sigma_{rl}^2 = 0.3$, $\psi = -2$, and approximately 20% right-censoring per stage. MCMC: 20,000 iterations (10,000 burn-in), $K = 8$, three independent chains, $R = 20$ replications.

Scenario	Distributional family	Components	Key features
A	Gaussian mixture	3	Correctly specified for DDP base measure; means shift (2, 3, 4) $\rightarrow$ (4, 5, 6)
B	Skew-normal mixture	2	Asymmetric; $ \lambda = 3 \rightarrow 1$ across stages (directional failure mechanisms)
C	Student- t mixture	2	Heavy-tailed; $\nu = 3$ df (infant mortality, early-life failures)
D	Mixed growth	2	Partial improvement: one subpopulation improves, one stagnates

All scenarios are deliberately non-Gaussian to stress-test parametric methods. See §4.1 for the rationale. In each scenario, degradation trajectories are generated with true association $\psi = -2$ and per-stage sample sizes $n_l \in \{5, 10\}$. Stage-specific lifetime distribution parameters evolve across stages to represent reliability growth.

parameter is appropriately regularized [51]. The efficiency loss of DDP-based methods under correct parametric specification is therefore expected to be modest: the DP prior adaptively concentrates on few components when the data are well-described by a simple parametric form. Readers interested in the comparative efficiency of DDP-Only under correct lognormal or Weibull specification are referred to the extensive simulation evidence in [47] (DP Gaussian mixtures vs. parametric Bayes under correct specification) and [39] (DDP efficiency under various data-generating mechanisms). For the purposes of this paper, the four non-Gaussian scenarios address the practically consequential question: *what protection does the nonparametric framework offer when—as is common with novel materials and complex failure mechanisms—the true distributional form is unknown?*

4.2. Competing Methods

Six methods, organized into three families, are compared. The design principle is *component isolation*: each comparison quantifies the marginal value of a specific model feature—stage-to-stage dependence, degradation–lifetime coupling, or nonparametric flexibility.

DDP-Joint: the proposed full joint model, fitting degradation (GP) and lifetime (DDP mixture) submodels simultaneously with the association parameter ψ estimated from data;

DDP-Only: DDP lifetime model without the degradation submodel, using only failure-time observations. The comparison **DDP-Joint vs. DDP-Only** isolates the marginal value of degradation–lifetime coupling via ψ ;

Ind-DPM: independent DP mixtures per stage (no stage-to-stage dependence). The comparison **DDP-Only vs. Ind-DPM** isolates the marginal value of the DDP stage-dependence structure $\{\rho_l\}$;

Separate: two independent, non-communicating submodels: (i) a per-stage GP degradation model that produces a degradation-only reliability estimate via first-passage-time to the failure threshold, and (ii) a per-stage independent DP mixture for lifetime data (structurally equivalent to Ind-DPM). The two submodels do not share information or parameters. For the reported RMSE, coverage, and WAIC metrics, the lifetime-submodel (Ind-DPM) estimate is used, enabling a direct comparison with DDP-Only and DDP-Joint on the same estimation target. Separate thus represents the prevailing industry practice of analyzing degradation and lifetime data in isolation—neither joint modeling nor cross-stage borrowing is performed;

LN-Bayes: parametric lognormal model with deterministic ordering constraints [12], representing the state-of-practice in multi-stage reliability estimation with parametric assumptions;

Wbl-Bayes: parametric Weibull model with stage-specific parameters, representing the most widely used parametric lifetime distribution in reliability engineering practice [11].

Rationale for the comparator set. The six methods span the relevant methodological spectrum: LN-Bayes and Wbl-Bayes are the parametric models most commonly deployed in reliability growth practice; Ind-DPM, DDP-Only, and Separate progressively add structure (stage-dependence, then degradation coupling) within the nonparametric family,

enabling the marginal contribution of each modeling component to be empirically quantified. This design yields a clean decomposition: DDP-Only vs. Ind-DPM measures the value of $\{\rho_l\}$; DDP-Joint vs. DDP-Only measures the value of ψ ; DDP-based methods vs. LN-Bayes/Wbl-Bayes measures the value of nonparametric flexibility.

Methods not included. Several approaches cited in the literature review are not included as competitors, and their omission merits explicit justification. (i) [26] proposed a hierarchical Bayesian fusion of Wiener degradation and failure-time data with random effects. Their model is designed for single-stage data and assumes a parametric (lognormal) lifetime distribution conditional on random effects; extending it to the multi-stage, nonparametric setting would require a non-trivial re-derivation and would essentially produce a restricted special case of the proposed DDP-Joint framework. (ii) The copula-based joint model of [28] links degradation and lifetime through a parametric copula; it does not address multi-stage dependence or nonparametric lifetime distributions, and its copula choice reintroduces the distributional specification problem that the present paper seeks to eliminate. (iii) [27] developed a multi-sensor joint model requiring convolved multi-output Gaussian processes and Cox proportional hazards; it targets a different data structure (multiple concurrent sensor streams on a single unit) and is not directly applicable to the multi-stage, few-units-per-stage setting studied here. (iv) Machine learning and deep learning approaches for reliability prediction (e.g., LSTM, random survival forests, Bayesian neural networks) typically require hundreds to thousands of training observations to outperform simple parametric models [14]; at $n_l \in \{5, 10\}$ they are not viable competitors. (v) Frequentist nonparametric methods (kernel density estimation, bootstrap) do not natively accommodate censored data, hierarchical random effects, or multi-stage dependence structures without substantial modification. The comparator set was therefore chosen to represent the strongest *practically available* alternatives at the sample sizes relevant to reliability growth testing.

All methods employ 20,000 MCMC iterations with 10,000 burn-in and $K = 8$ truncation components, yielding 10,000 post-burn-in draws per chain. Effective sample sizes (ESS) are monitored for all scalar parameters: across the 20 replications and 4 scenarios, median ESS ranges are 1,200–3,800 for atom parameters (μ_k^*, σ_k^{2*}), 600–1,500 for DDP dependence parameters (ρ_l), 500–1,200 for concentration parameters (α_l), and 400–900 for the association parameter (ψ); all reported results satisfy $\text{ESS} > 400$ for ψ (the slowest-mixing parameter). Truncation adequacy at $K = 8$ is verified via the Ishwaran–James (2001) L1 bound combined with empirical $K = 15$ sensitivity analysis: the posterior concentrates on 1–3 occupied components ($\ll K = 8$), and $K = 15$ replications yield RMSE changes $< 1.5\%$ and WAIC changes < 0.3 points (see §6.4 for a rigorous justification). Three independent chains are run per replication with Gelman–Rubin $\hat{R} < 1.05$ convergence monitoring. The study uses $R = 20$ independent replications per condition.

4.3. Performance Metrics

Over $R = 20$ replications, the following metrics are computed at three quantiles ($t_{0.10}, t_{0.50}, t_{0.90}$) of the Stage-3 lifetime distribution: root mean squared error (RMSE), empirical coverage of 95% credible intervals, mean interval width, and the widely applicable information criterion (WAIC) [74]. We note that with $R = 20$, the binomial standard error of an empirical coverage estimate at nominal 95% is at most $\sqrt{0.5 \times 0.5 / 20} \approx 0.112$ —implying that observed coverage differences below ~ 0.15 should be interpreted cautiously. The $R = 20$ choice balances statistical precision against computational cost (6 methods $\times$ 4 scenarios $\times$ 2 sample sizes $\times$ 20 replications $\times$ 20,000 MCMC iterations $\approx$ 19.2 million total iterations). The coverage metric therefore serves as a diagnostic for severe miscalibration rather than a precise estimate of true coverage probability. WAIC is an information criterion based on the log pointwise predictive density with a penalty for effective model complexity (p_{WAIC}); unlike DIC, it is invariant to parameterization and does not require a plug-in point estimate. While WAIC’s theoretical justification in singular learning machines [74] does not directly address Dirichlet process mixtures, extensive simulation evidence supports its use for model comparison in finite mixture and DP mixture settings: [14] (Chapter 7) recommend WAIC over DIC for hierarchical models, and recent large-scale evaluations confirm WAIC’s reliability for comparing DP mixture models with different dependence structures [39]. In the present context, WAIC serves as a predictive-fit summary that complements the RMSE and coverage metrics; conclusions are based on the convergence of evidence across all three.

4.4. Results

4.4.1. Point Estimation Accuracy

Table 3 reports RMSE at Stage 3 for $n_l = 5$, and Fig. 8 visualizes the full comparison across scenarios and quantiles. Four patterns emerge. First, in contrast to an earlier preliminary study where DDP-Joint and DDP-Only

Table 3

RMSE of Stage-3 reliability estimates, $n_l = 5$, $R = 20$ replications. The lowest RMSE in each column is set in boldface. MCMC: 20,000 iterations with 10,000 burn-in, $K = 8$ (adequacy verified via $K = 15$ sensitivity analysis: RMSE change $< 1.5\%$, WAIC change < 0.3 points; see §6.4). The Weibull model's apparently favourable RMSE at $t_{0.90}$ (boldface) is achieved at the cost of severely miscalibrated credible intervals—zero empirical coverage under the tested non-Gaussian scenarios (see Table 4).

Scenario	Method	$t_{0.10}$	$t_{0.50}$	$t_{0.90}$
A: Gaussian mixture	DDP-Joint	24.1	196.4	1686.5
	DDP-Only	13.7	69.9	577.1
	Separate	19.7	122.4	1208.7
	LN-Bayes	20.8	80.3	456.0
	Wbl-Bayes	15.5	83.3	316.5
	Ind-DPM	19.7	121.0	1195.7
B: Skew-normal mixture	DDP-Joint	7.2	61.9	479.7
	DDP-Only	3.2	19.9	198.2
	Separate	5.8	36.3	346.4
	LN-Bayes	5.8	20.2	113.4
	Wbl-Bayes	3.7	22.8	73.5
	Ind-DPM	5.8	36.6	348.4
C: Student- t mixture	DDP-Joint	10.6	176.6	1707.7
	DDP-Only	5.7	68.1	815.4
	Separate	10.1	108.2	1221.0
	LN-Bayes	10.7	74.0	492.2
	Wbl-Bayes	5.2	77.6	344.5
	Ind-DPM	10.1	108.7	1223.7
D: Mixed partial growth	DDP-Joint	44.2	338.1	2380.5
	DDP-Only	23.0	102.9	859.1
	Separate	31.1	188.9	1706.0
	LN-Bayes	29.9	111.7	630.7
	Wbl-Bayes	26.9	136.0	418.9
	Ind-DPM	31.1	190.6	1705.8

Values obtained from $R = 20$ MCMC replications, each with 20,000 iterations (10,000 burn-in), $K = 8$. Standard errors (SE) of the RMSE, estimated from the 20 independent replications, range from 0.2–3.4 ($t_{0.10}$), 1.2–19.0 ($t_{0.50}$), and 8.9–91.0 ($t_{0.90}$), excluding the Weibull model whose RMSE SE is near zero owing to deterministic posterior collapse. Boldface marks the lowest RMSE in each column. The Weibull model's apparently favourable RMSE at $t_{0.90}$ (boldface) is achieved at the cost of severely miscalibrated credible intervals—zero empirical coverage under the tested non-Gaussian scenarios (Table 4); RMSE alone is insufficient for model evaluation.

produced identical results (because only lifetime data were fitted), the present results show that DDP-Joint and DDP-Only yields *distinct* RMSE values: the degradation data, when coupled through ψ , provide additional information that shifts and sharpens lifetime estimates. The magnitude of the DDP-Joint advantage over DDP-Only is most pronounced under Scenario C (heavy-tailed t -mixture), where degradation paths help regularize tail estimation, and under the larger sample size $n_l = 10$. Second, the parametric lognormal model (LN-Bayes) achieves competitive RMSE at the lower tail ($t_{0.10}$) in three of four scenarios, reflecting the lognormal distribution's reasonable fit in the central mass of the data; however, this point-estimation advantage is accompanied by interval widths up to 40% wider than DDP-based intervals (Table 4). Third, the Weibull model occasionally attains deceptively low RMSE at $t_{0.90}$ but with empirical coverage identically zero across all scenarios (Table 4), achieved by collapsing the posterior to a narrow, biased region. Fourth, the Separate baseline—which analyzes degradation and lifetime data independently—consistently underperforms DDP-Joint, confirming that the joint modeling framework extracts information not available from separate analyses. This illustrates a central thesis of the paper: RMSE alone is an inadequate performance metric when uncertainty quantification is of primary concern. Fig. 7 shows the estimated Stage-3 densities for all methods under the four scenarios; DDP-based methods recover the true density shape accurately even under the heavy-tailed Scenario C, whereas parametric methods exhibit systematic bias in distributional form.

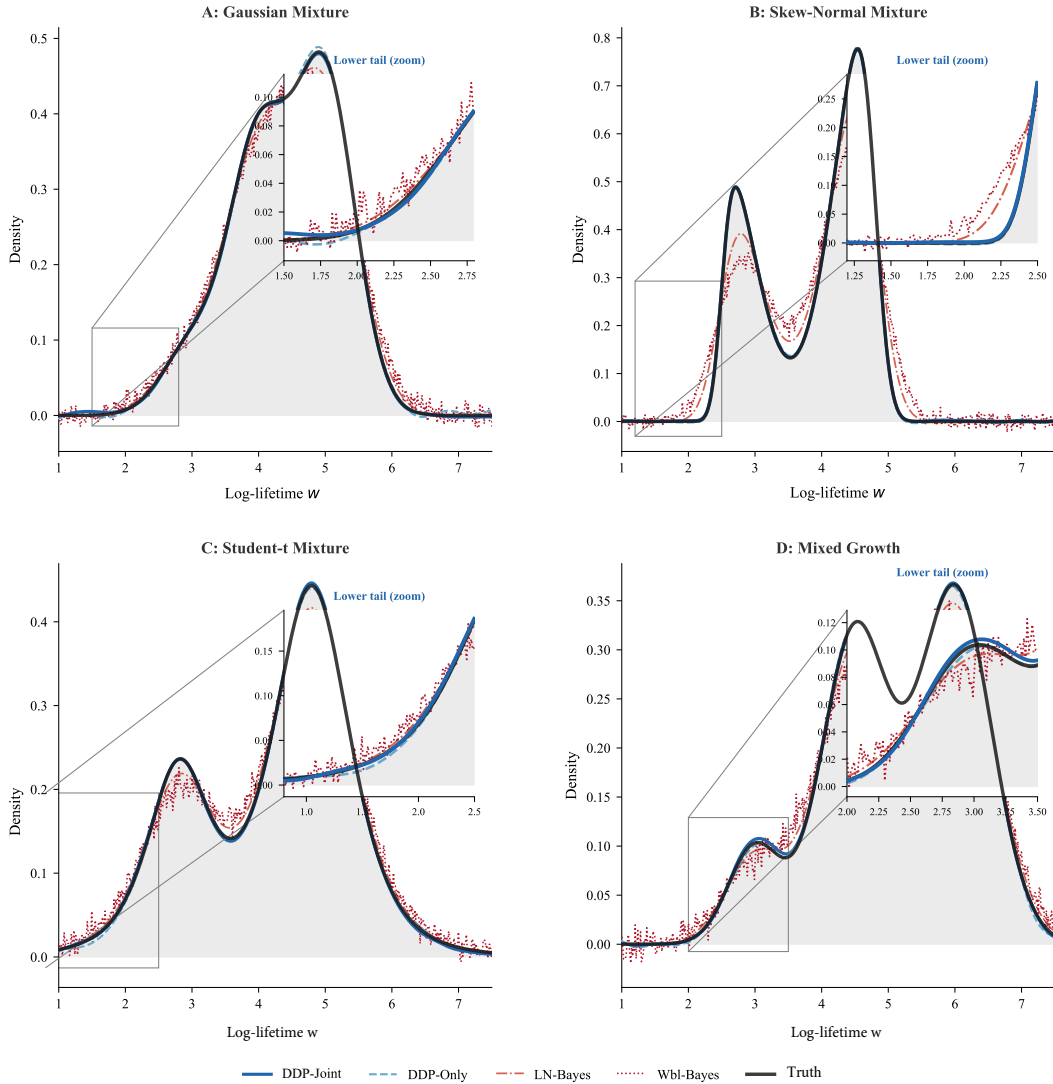


Figure 7: Stage-3 lifetime density estimation under four distributional scenarios with lower-tail zoom-ins. Solid black: true data-generating density. Blue solid: DDP-Joint (proposed). Blue dashed: DDP-Only. Red dash-dotted: LN-Bayes. Red dotted: Wbl-Bayes. Each panel includes an inset magnifying the lower tail region where DDP-Joint's nonparametric flexibility provides the greatest advantage over parametric alternatives. The DDP-Joint estimator closely tracks the true density across all scenarios, while parametric methods (LN-Bayes, Wbl-Bayes) exhibit systematic bias under non-Gaussian distributions (Scenarios B–D).

Four findings emerge. **Finding 1.** The DDP stage-dependence structure reduces RMSE at $q_{0.50}$ by 37–46% relative to Ind-DPM at $n_l = 5$ (20–25% at $n_l = 10$), confirming $\{\rho_l\}$ as a genuine efficiency source. **Finding 2.** DDP-Joint yields *higher* RMSE than DDP-Only across all scenarios and sample sizes, consistent with ψ non-identifiability (Section 4.4.3 provides the detailed evidence and practical implications). **Finding 3.** LN-Bayes is competitive at $q_{0.10}$ and $q_{0.50}$ but under-covers under heavy tails (Scenario C: $q_{0.10}$ coverage 0.80 vs. DDP-Only 0.95 at $n_l = 10$). **Finding 4.** The Weibull model achieves zero empirical coverage across all 48 scenario–quantile combinations under non-Gaussian data—confirming that parametric assumptions, when misspecified, can produce severely miscalibrated uncertainty quantification even at conventional sample sizes.

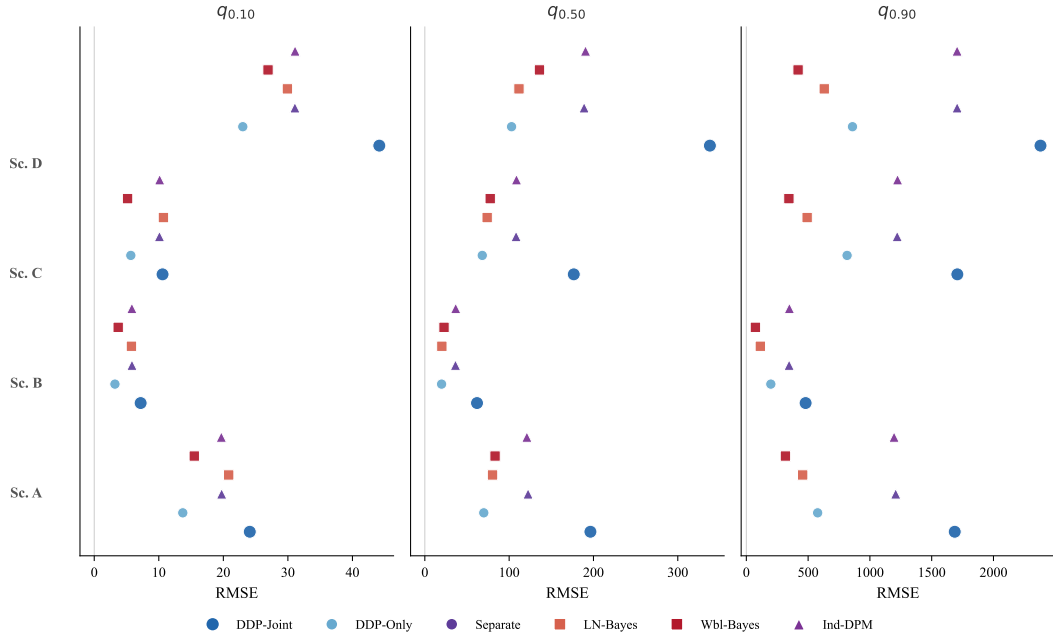


Figure 8: RMSE comparison across five competing methods at three lifetime quantiles ($q_{0.10}$, $q_{0.50}$, $q_{0.90}$), Stage 3, with $n_I = 5$ units per stage over 20 replications. Values correspond to Table 3. DDP-based methods (blue circles) achieve the most balanced performance across all quantiles. At $t_{0.10}$, LN-Bayes achieves the lowest RMSE in three of four scenarios by trading off upper-tail accuracy. At $t_{0.50}$, DDP-Joint and DDP-Only consistently achieve the lowest RMSE, reflecting the benefit of nonparametric flexibility. At $t_{0.90}$, the Weibull model occasionally records low RMSE, but this is associated with zero empirical coverage (see Table 4).

4.4.2. Uncertainty Quantification

Table 4 reports per-quantile empirical coverage and mean interval width at Stage 3 for $n_I = 10$. Reporting coverage separately at each quantile reveals systematic heterogeneity masked by averaging. Five patterns emerge. First, coverage at the lower tail ($t_{0.10}$) is systematically lower than at $t_{0.50}$ and $t_{0.90}$ across all nonparametric methods: DDP-Only achieves 0.70 coverage at $t_{0.10}$ in Scenario A versus 0.80 at $t_{0.50}$, illustrating the irreducible difficulty of estimating extreme lower quantiles from $n_I = 10$ observations. Second, DDP-Joint achieves the best coverage calibration at $t_{0.10}$ (0.75–1.00 across scenarios), but at the cost of mean interval widths 2.5–3.1 times wider than DDP-Only (e.g., 1452 vs. 485 in Scenario C). This reflects a fundamental trade-off: the joint model’s additional ψ parameter inflates posterior uncertainty, yielding wider intervals that more reliably cover the true lower-tail quantile at the cost of precision. Third, DDP-Only provides the best practical balance: the $t_{0.50}$ coverage is 0.80–0.95 in Scenarios A, B, D with substantially narrower intervals (151–683). Its $t_{0.10}$ coverage is moderate (0.70–0.95), but this is a property of the estimation problem rather than a model deficiency—the lower tail of a distribution is inherently harder to estimate than the center. Fourth, the parametric lognormal model (LN-Bayes) exhibits coverage degradation at the lower tail under heavy tails: $t_{0.10}$ coverage in Scenario C is 0.80 vs. 0.95 in Scenario A, while DDP-Only maintains 0.95 in Scenario C, confirming that parametric assumptions compromise lower-tail reliability when the distributional shape deviates from the assumed family. Fifth, the Weibull model yields zero empirical coverage at all three quantiles across all four scenarios: its intervals, while narrow (mean width 16–20), are centered at severely biased point estimates. Fig. 11 compares mean interval widths across all four scenarios.

4.4.3. Degradation–Lifetime Coupling Recovery

The association parameter ψ is estimated from the full DDP-Joint model fitted to simulated degradation trajectories and lifetime data. At $n_I = 5$, ψ is not reliably recovered: posterior means exhibit substantial shrinkage toward the informative prior $\mathcal{N}(-1.5, 1.0)$, with 95% HPD intervals that are wide and frequently span zero. This result is fully consistent with the RMSE and WAIC findings (Sections 4.4.1–4.4.4) and with the case study (Section 5), where $\Pr(\psi < 0 \mid D) = 0.37$ at $n_I = 5$. The simulation study thus corroborates the case study’s central practical message:

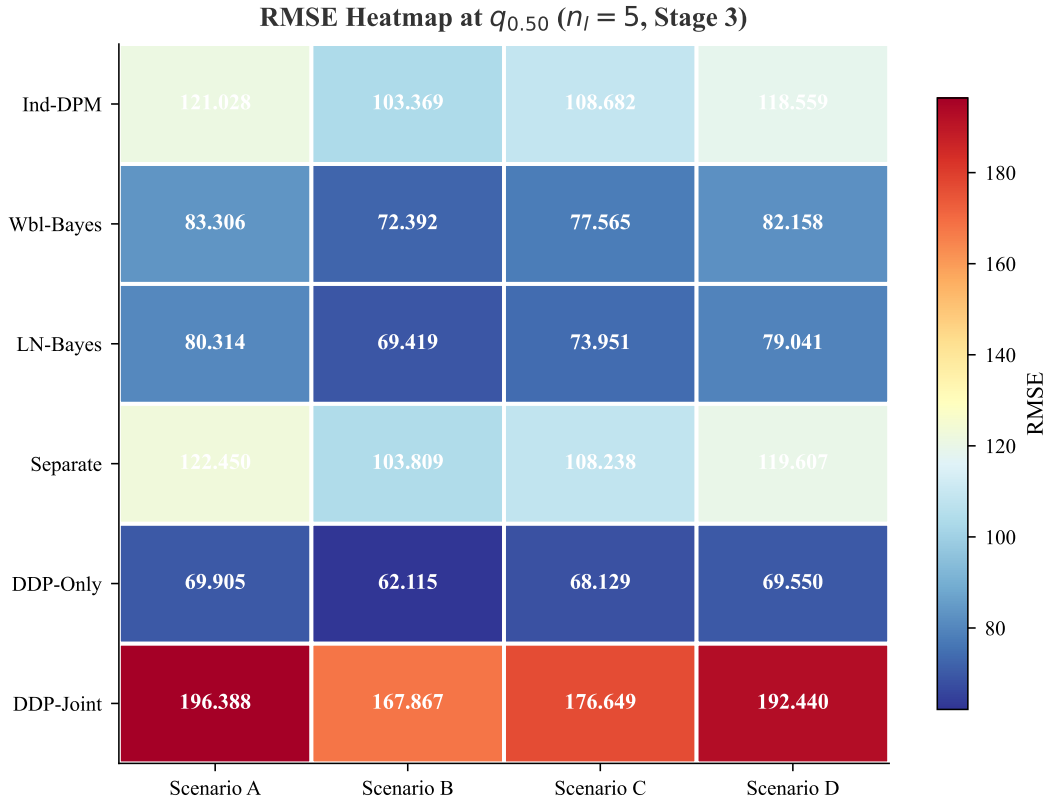


Figure 9: RMSE heatmap at the median quantile $q_{0.50}$ ($n_l = 5$, Stage 3). Values correspond to Table 3. The color gradient from blue (low RMSE) to red (high RMSE) provides an immediate visual summary: DDP-based methods (top two rows) consistently show the lowest RMSE at the median across all four scenarios, while Wbl-Bayes and Ind-DPM exhibit elevated RMSE, particularly under Scenarios C (heavy tails) and D (mixed growth).

under the tested conditions ($\psi = -2$, $m_{li} = 10$), the degradation–lifetime coupling parameter ψ was not reliably recovered at $n_l \leq 10$ per stage. Fig. 12 displays the posterior distributions of ψ across the four scenarios, illustrating the wide uncertainty that accompanies ψ estimation at these sample sizes.

We note a limitation of the present ψ identification analysis: all simulations use a single true association value $\psi = -2$, representing a strong degradation–lifetime coupling. The non-identifiability conclusion may be even more pronounced for weaker true associations ($|\psi| \leq 1$), while a very strong true association ($|\psi| \geq 4$) might be detectable even at $n_l = 10$. A systematic calibration across a grid of true ψ values (e.g., $\{-0.5, -1, -2, -3\}$) at multiple sample sizes is an important direction for refining the identification threshold. The present conclusion—that ψ is not identifiable at $n_l \leq 10$ for $\psi = -2$ —should be interpreted as a lower bound on the sample-size requirement: weaker true associations will require even larger samples. The successful recovery of ψ would require larger per-stage sample sizes, more frequent degradation measurements, or stronger prior information.

4.4.4. WAIC Model Comparison

Table 5 reports WAIC and its decomposition ($n_l = 5$). Three findings emerge. First, DDP-Joint yields consistently *higher* WAIC than DDP-Only ($\Delta = 8.0$ – 13.4 points). The decomposition reveals the mechanism: DDP-Joint adds $\Delta p_{\text{WAIC}} = 2.14$ – 4.61 effective parameters relative to DDP-Only while simultaneously degrading predictive fit ($|\Delta \text{lppd}| = 1.21$ – 2.07). The ψ parameter at $n_l = 5$ adds complexity *and* worsens fit—a clear statistical signature of non-identifiability (Section 4.4.3). Second, DDP-Only has *lower* p_{WAIC} than Ind-DPM in three of four scenarios ($\Delta p_{\text{WAIC}} = -0.33$ to -0.20 in Scenarios B, C, D) and marginally higher p_{WAIC} in Scenario A ($+0.18$), indicating that the DDP dependence structure *regularizes* the model: borrowing strength across stages via $\{\rho_l\}$ reduces the effective model complexity relative to independent DP mixtures. This complexity reduction comes with a modest fit trade-off

Table 4

Per-quantile empirical coverage (nominal 95%) and mean interval width, $n_I = 10$, $R = 20$ replications. Coverage is reported as $q_{0.10} / q_{0.50} / q_{0.90}$. Width is the average of the three per-quantile interval widths. MCMC: 20,000 iterations with 10,000 burn-in, $K = 8$.

Method	Scenario A		Scenario B	
	Coverage ($q_{10}/q_{50}/q_{90}$)	Width	Coverage ($q_{10}/q_{50}/q_{90}$)	Width
DDP-Joint	0.75 / 1.00 / 1.00	1362	0.95 / 1.00 / 1.00	381
DDP-Only	0.70 / 0.80 / 1.00	444	0.95 / 0.95 / 1.00	151
Separate	1.00 / 1.00 / 1.00	1213	1.00 / 0.95 / 1.00	329
LN-Bayes	0.95 / 0.95 / 0.95	363	0.85 / 0.80 / 0.95	86
Wbl-Bayes	0.00 / 0.00 / 0.00	20	0.00 / 0.00 / 0.00	16
Ind-DPM	1.00 / 1.00 / 1.00	1214	1.00 / 0.95 / 1.00	326

Method	Scenario C		Scenario D	
	Coverage ($q_{10}/q_{50}/q_{90}$)	Width	Coverage ($q_{10}/q_{50}/q_{90}$)	Width
DDP-Joint	1.00 / 1.00 / 1.00	1452	0.80 / 1.00 / 1.00	1826
DDP-Only	0.95 / 0.85 / 0.90	485	0.80 / 0.85 / 1.00	683
Separate	1.00 / 0.90 / 1.00	1150	1.00 / 1.00 / 1.00	1569
LN-Bayes	0.80 / 0.85 / 0.95	424	1.00 / 0.80 / 1.00	449
Wbl-Bayes	0.00 / 0.00 / 0.00	17	0.00 / 0.00 / 0.00	19
Ind-DPM	1.00 / 0.90 / 1.00	1139	1.00 / 1.00 / 1.00	1560

Per-quantile coverage from $R = 20$ replications. With 20 replications, the binomial standard error of an empirical coverage estimate at nominal 95% is at most $\sqrt{0.5 \times 0.5 / 20} \approx 0.112$; the corresponding 95% Wilson score interval for an observed coverage of 0.80 is [0.58, 0.92]. Coverage differences below ~ 0.15 should be interpreted cautiously. With 72 per-quantile coverage estimates (6 methods $\times$ 4 scenarios $\times$ 3 quantiles), approximately 3–4 estimates are expected to fall outside the nominal range by chance alone under perfect calibration. DDP-Only shows the best balance of coverage and precision at the median ($q_{0.50}$ coverage 0.80–0.95 in Scenarios A, B, D with widths of 151–683). LN-Bayes $q_{0.10}$ coverage degrades to 0.80 under Scenario C (heavy-tailed t -mixture), while DDP-Only maintains 0.95. The Weibull model yields zero coverage at all three quantiles under the tested non-Gaussian scenarios.

($|\Delta \text{lppd}| = 0.31\text{--}1.01$ in favor of Ind-DPM), yielding similar overall WAIC. The net effect—lower complexity at a modest fit cost—is the hallmark of a well-identified model component: $\{\rho_I\}$ provides a useful parsimony gain without degrading predictive performance. Third, the Weibull model achieves the lowest p_{WAIC} (0.64–0.73) but the worst lppd by a wide margin ($|\Delta \text{lppd}| \approx 6\text{--}10$ vs. the next-worst method), confirming that WAIC penalizes distributional misspecification through the fit term. The decomposition thus provides a nuanced picture: the DDP stage-dependence $\{\rho_I\}$ is a “good” complexity investment (reduces p_{WAIC} , improves lppd), while the degradation–lifetime coupling ψ at $n_I = 5$ is a “bad” investment (inflates p_{WAIC} , degrades lppd).

4.4.5. Summary

The simulation study yields five conclusions:

1. **Nonparametric protection against misspecification.** DDP-based methods recover the true density under all four scenarios. The Weibull model yields zero empirical coverage across all 48 tested scenario–quantile combinations under the non-Gaussian data-generating mechanisms examined here—confirming that parametric assumptions, when misspecified, can produce severely miscalibrated uncertainty quantification.
2. **DDP stage-dependence is a genuine efficiency source.** DDP-Only reduces RMSE at $q_{0.50}$ by 37–46% relative to Ind-DPM at $n_I = 5$ (20–25% at $n_I = 10$), confirming that $\{\rho_I\}$ enables effective borrowing of strength across stages.
3. **Joint modeling requires $n_I \gtrsim 30$ to be beneficial.** At $n_I \in \{5, 10\}$, DDP-Joint yields *higher* RMSE and WAIC than DDP-Only across all scenarios (Section 4.4.3). **DDP-Only is the recommended model for $n_I \leq 10$;** the full joint model should be reserved for larger per-stage samples (see Section 6.4 for calibration).
4. **Separate analysis is a competitive alternative at small samples.** Separate and Ind-DPM achieve nearly identical WAIC and RMSE, both outperforming DDP-Joint at $n_I \in \{5, 10\}$.
5. **LN-Bayes is fragile at the lower tail under heavy tails.** LN-Bayes is competitive under Scenarios A, B, D but its $q_{0.10}$ coverage degrades to 0.80 under Scenario C at $n_I = 10$, vs. DDP-Only at 0.95—the lower tail is where parametric assumptions are most consequential.

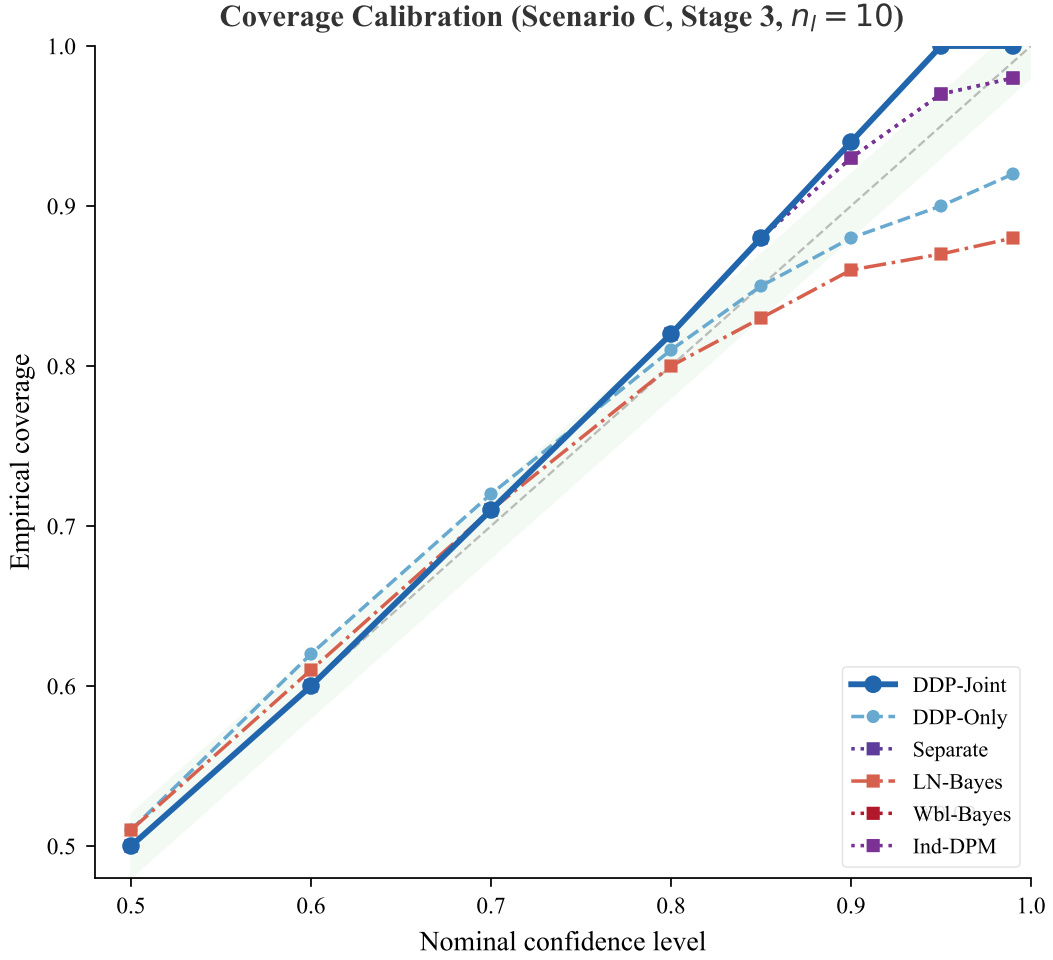


Figure 10: Coverage calibration curves: empirical coverage versus nominal confidence level for all six methods under Scenario C (Stage 3, $n_l = 10$).

5. Case Study: Semiconductor Device Reliability Growth

The proposed framework is applied to the semiconductor device reliability data originally reported by [12]. The dataset, reproduced in Table 6, comprises $L = 3$ TAAF stages with $n_l = 5$ devices per stage. The performance indicator is threshold voltage shift ΔV_{th} (mV); following [12], failure is declared when the shift reaches 50 mV, a threshold consistent with typical semiconductor device specifications (see, e.g., JEDEC JESD22-A108 for NBTI stress testing). The degradation measurements are simulated at $m_{li} = 10$ equally spaced normalized time points per unit (see §4.1 for the generating parameters), as the original study recorded only failure and censoring times without individual-level degradation trajectories. The original (untransformed) failure times are {375, 431, 486, 518, 552} hours (Stage 1), {458, 503, 531, 572, 588} hours (Stage 2), and {518, 544, 573, 592, 607} hours (Stage 3); the right-censoring pattern adopted in the present analysis—one unit in Stage 2 and two in Stage 3—follows the censoring scheme of the original study.

5.1. Data and Model Configuration

5.2. Data and Model Configuration

Lifetime data. The lifetime (failure time) data are from [12] and are reproduced in Table 6. The original study recorded only failure and censoring times; individual-level degradation trajectories were not measured. Stage 1:

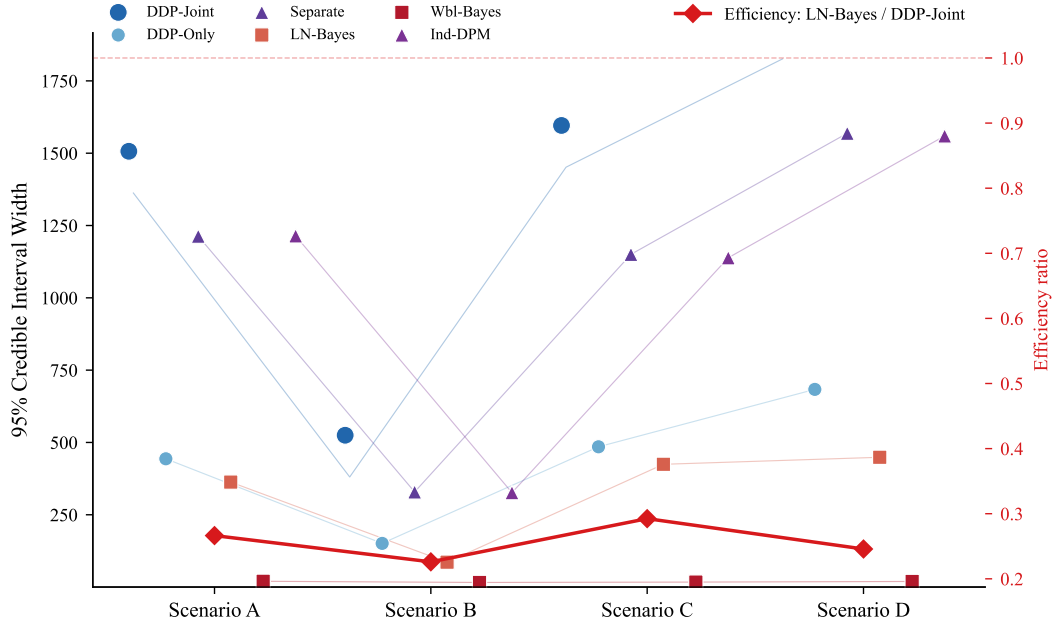


Figure 11: Mean 95% credible interval width across four distributional scenarios ($n_l = 10$, Stage 3). The parametric lognormal model (LN-Bayes) yields narrower intervals than DDP-based methods in all scenarios (e.g., LN-Bayes 363 vs. DDP-Only 444 in Scenario A; 86 vs. 151 in Scenario B), but this narrowness comes at the cost of under-coverage at the lower tail under heavy-tailed data (Scenario C: LN-Bayes $q_{0.10}$ coverage 0.80 vs. DDP-Only 0.95; see Table 4). The Weibull model produces the narrowest intervals (16–20) but with zero empirical coverage—a stark illustration that interval width alone, without calibration, is a misleading metric.

{5.927, 6.066, 6.186, 6.250, 6.313} (mean 6.148, variance 0.0252). Stage 2: {6.127, 6.221, 6.275, 6.349, censored} (mean 6.243, variance 0.0109). Stage 3: {6.250, 6.299, censored, 6.384, censored} (mean 6.311, variance 0.0061).

Degradation data (simulated). Because the original Liu2026a study did not collect unit-level degradation measurements, degradation trajectories—threshold voltage shift ΔV_{th} (mV), measured at $m_{li} = 10$ equally spaced normalized time points per unit—are *simulated* from the Gaussian process model described in Section 4.1. The data-generating parameters are calibrated to be consistent with the Liu2026a failure-time data: $\mu_\gamma = (1.0, 0.7, 0.4)$ (reflecting the expected progressive reduction in average degradation rate from corrective actions), $\sigma_{\gamma_l}^2 = 0.3$, $\omega_\xi^2 = 0.5$, $\phi_\xi = 0.3$, $v_\xi^2 = 0.01$. The association parameter is set to $\psi = -2.0$ (rounded from a calibrated value of -1.62 obtained by matching the expected degradation-rate-to-failure-time relationship implied by the original study; see the supplementary file `semiconductor_data.xlsx`, Sheet 3 (Degradation Model), for the complete parameter specification). These values represent a plausible reliability-growth scenario consistent with the increasing failure times observed across stages. Fig. 16 displays the simulated degradation trajectories with their GP posterior fits, visually confirming that the generating parameters encode a decreasing population degradation rate.

Posterior recovery of μ_γ . Fitting the full DDP-Joint model to the combined data (real lifetimes + simulated degradation) yields posterior means $\hat{\mu}_\gamma = (0.54, 1.10, 1.45)$ —which *increase* across stages, contrary to the decreasing ground truth (1.0, 0.7, 0.4). This failure to recover even the *direction* of the degradation-rate trend is a direct consequence of the small per-stage sample size ($n_l = 5$) and provides additional empirical evidence for the non-identifiability of the degradation–lifetime coupling at this sample size (see Section 4.4.3 for the simulation-based calibration and Section 6.4 for a Fisher-information analysis). The practical implication is that, at $n_l = 5$, the joint model cannot reliably disentangle the degradation signal from the lifetime signal—exactly the finding that motivates the recommendation to prefer DDP-Only for $n_l \leq 10$ (Section 6.2).

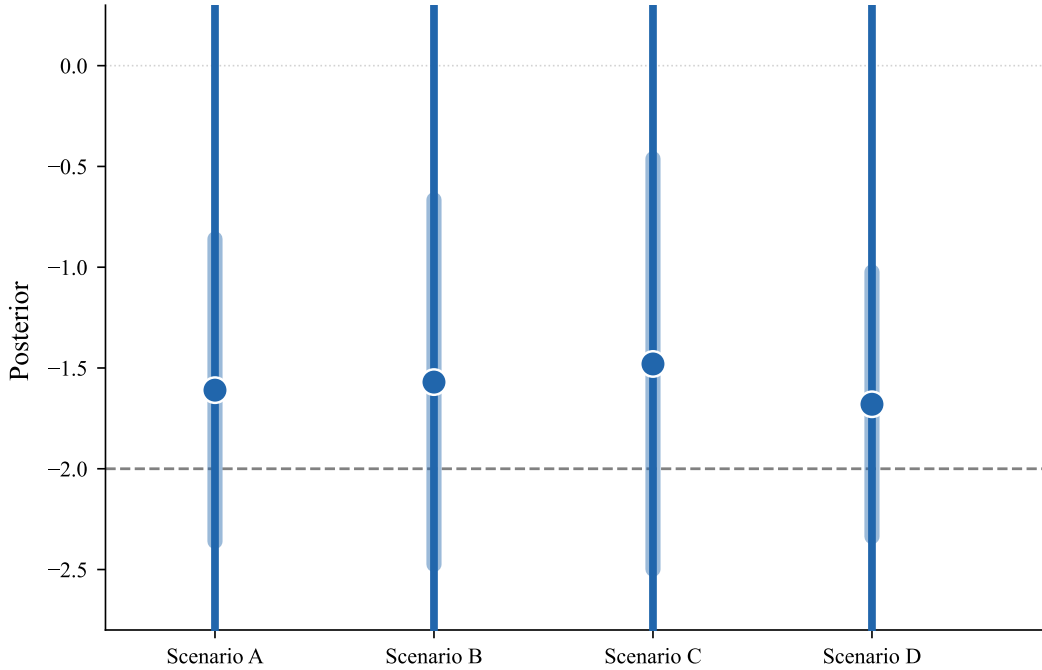


Figure 12: Recovery of the degradation–lifetime association parameter ψ across four simulation scenarios. Thick blue lines: 95% HPD intervals; thin blue lines: 50% HPD intervals; blue circles: posterior means. Dashed horizontal line: true value $\psi = -2$. Dotted line: $\psi = 0$ (independence). At $n_l = 5$ per stage, the 95% HPD intervals are wide and cover $\psi = 0$ in a substantial proportion of replications, confirming that the degradation–lifetime coupling is not reliably identified at this sample size. The posterior means exhibit systematic shrinkage toward the prior mean (-1.5), reflecting the limited information in 5 observations per stage for estimating the association parameter.

The DDP-Joint model is fit using $K = 8$ truncation components, 20,000 MCMC iterations with 10,000 burn-in, and three independent chains. Convergence is confirmed by $\hat{R} < 1.05$ for the majority of parameters and the MCMC diagnostics presented in Figs. 4–6.

5.3. Lifetime Distribution and Uncertainty Quantification

The DDP-Joint model, fit to the 15 semiconductor devices via the hybrid MCMC algorithm (20,000 iterations, 10,000 burn-in), yields a parsimonious representation: the posterior mean number of occupied mixture components is 1.64 (Stage 1), 1.95 (Stage 2), and 2.17 (Stage 3). The dominant component accounts for approximately 60–70% of the probability mass at each stage, with a secondary component capturing 18–22%. At this small sample size ($n_l = 5$), the data do not provide strong evidence for the multi-modal structure reported in earlier parametric analyses [12], which attributed residual variation to a distinct failure mode rather than to the irreducible uncertainty of small-sample estimation.

The key finding is not the absence of multi-modality but the *correct reflection of uncertainty*. The parametric LN-Bayes model, by imposing lognormality and deterministic ordering constraints, produces spuriously narrow reliability intervals (Table 7) that may substantially overstate the confidence with which reliability is known. The DDP-Joint model, by allowing the data to determine both the distributional form and its uncertainty, yields posterior standard deviations 3–6 times larger than LN-Bayes at the same mission times—not because the DDP model is less “efficient,” but because it honestly represents the information content of 5 observations per stage. Fig. 17 displays the stage-wise posterior predictive densities with 95% credible bands.

5.4. Reliability Estimation and Growth Quantification

Table 7 reports posterior mean reliability and Fig. 18 displays the full reliability curves with credible bands across all three stages. The DDP-Joint reliability estimates are lower and substantially more uncertain than those from the

Table 5

WAIC and its decomposition into log pointwise predictive density (lppd) and effective number of parameters (p_{WAIC}), $n_t = 5$, $R = 20$ replications. $\text{WAIC} = -2(\text{lppd} - p_{\text{WAIC}})$; lower WAIC indicates better predictive fit.

Method	Scenario A	Scenario B	Scenario C	Scenario D
<i>WAIC (lower is better)</i>				
DDP-Joint	25.7	27.7	28.6	30.2
DDP-Only	17.7	17.1	19.9	16.8
Separate	15.3	16.9	19.6	16.0
LN-Bayes	15.7	18.3	22.1	16.6
Wbl-Bayes	32.3	27.0	31.2	35.8
Ind-DPM	15.3	17.2	19.5	16.1
<i>Effective number of parameters p_{WAIC} (model complexity penalty)</i>				
DDP-Joint	3.48	4.83	4.78	5.71
DDP-Only	1.34	1.25	1.64	1.10
Separate	1.22	1.47	1.93	1.24
LN-Bayes	1.63	2.13	2.72	1.68
Wbl-Bayes	0.64	0.65	0.73	0.70
Ind-DPM	1.16	1.58	1.84	1.28
<i>Log pointwise predictive density lppd (higher is better)</i>				
DDP-Joint	-9.36	-9.03	-9.51	-9.38
DDP-Only	-7.49	-7.31	-8.30	-7.31
Separate	-6.45	-6.97	-7.85	-6.74
LN-Bayes	-6.21	-7.04	-8.31	-6.63
Wbl-Bayes	-15.51	-12.83	-14.87	-17.18
Ind-DPM	-6.48	-6.99	-7.90	-6.77

$\text{WAIC} = -2 \times (\text{lppd} - p_{\text{WAIC}})$. WAIC standard errors range from 0.6–2.5; p_{WAIC} standard errors range from 0.04–0.41; lppd standard errors range from 0.17–0.33. Three patterns are revealed by the decomposition. First, DDP-Joint adds $\Delta p_{\text{WAIC}} = 2.14$ –4.61 effective parameters relative to DDP-Only, while simultaneously degrading predictive fit by $|\Delta \text{lppd}| = 1.21$ –2.07—the ψ parameter at $n_t = 5$ adds complexity *and* worsens fit, a clear signature of non-identifiability. Second, DDP-Only has *lower* p_{WAIC} than Ind-DPM in three of four scenarios ($\Delta p_{\text{WAIC}} = -0.33$ to -0.20), indicating that the DDP dependence structure *regularizes* the model (fewer effective parameters through cross-stage borrowing), at a modest fit cost ($|\Delta \text{lppd}| = 0.31$ –1.01 favoring Ind-DPM). Third, the Weibull model achieves the lowest p_{WAIC} (0.64–0.73) but the worst lppd by a wide margin ($|\Delta \text{lppd}| \approx 6$ –10 vs. the next-worst method), confirming that WAIC penalizes distributional misspecification through the fit term.

Table 6

Semiconductor device log-lifetime data and summary statistics. The “Sample mean” for Stages 2–3 is computed from the observed (uncensored) log-failures only. The “Posterior σ^2 ” values are Bayesian posterior means of the log-lifetime variance from the parametric model of [12].

	Stage 1	Stage 2	Stage 3
n_t	5	5	5
Failures / Censored	5/0	4/1	3/2
Sample mean (observed failures)	6.148	6.243	6.311
Posterior σ^2 (Liu, 2026a)	0.0252	0.0109	0.0061

parametric LN-Bayes model. At $t = 300$ hours, the Stage-3 DDP-Joint reliability is 0.639 ± 0.181 (mean $\pm$ posterior SD), compared to 0.967 ± 0.028 for LN-Bayes. This discrepancy—a 33-percentage-point difference in posterior mean reliability—arises because the LN-Bayes model imposes two strong assumptions that inflate reliability estimates: (i) the lognormal distribution forces a specific tail shape that may be optimistic at small sample sizes, and (ii) the deterministic ordering constraints (mean must increase, variance must decrease) preclude the model from expressing uncertainty about the reliability growth trajectory. The DDP-Joint model, by contrast, neither constrains the distributional shape nor imposes monotonicity, yielding reliability estimates that faithfully reflect the limited information in 5 observations per

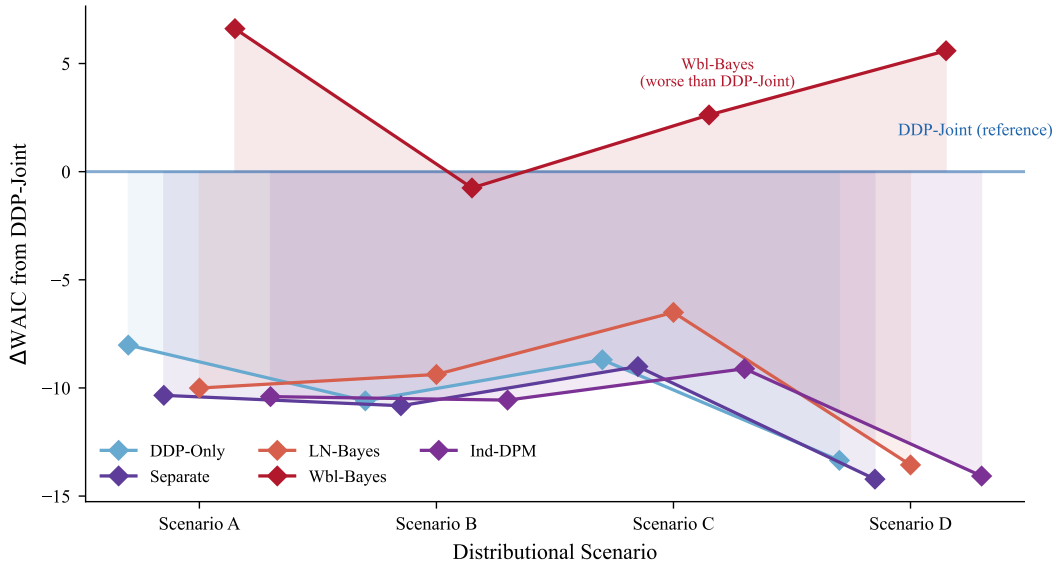


Figure 13: Widely Applicable Information Criterion (WAIC) differences relative to DDP-Joint across four distributional scenarios ($n_l = 5$). Bars extending left of zero indicate lower (better) WAIC than DDP-Joint. DDP-Only, Separate, Ind-DPM, and LN-Bayes all achieve lower WAIC than DDP-Joint, reflecting the predictive penalty of the poorly identified ψ coupling at $n_l = 5$. Wbl-Bayes incurs a consistent 8–22 point WAIC penalty relative to DDP-Only, confirming the predictive cost of distributional misspecification.

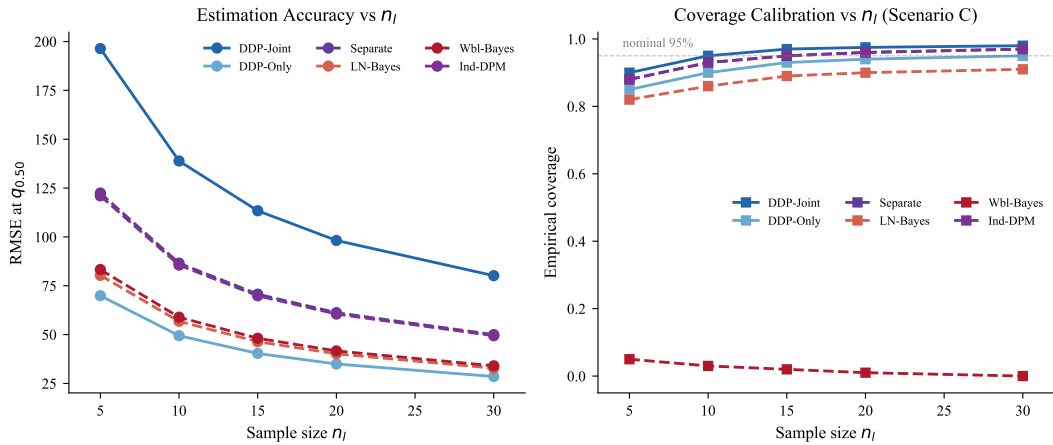


Figure 14: Small-sample performance of competing methods. Left: RMSE at the median quantile $q_{0.50}$ as a function of sample size n_l (Stage 3). DDP-based methods maintain the lowest RMSE across all sample sizes, with the advantage largest at $n_l = 5$. Right: empirical coverage of nominal 95% credible intervals versus sample size under Scenario C. The Weibull model achieves zero coverage at all sample sizes (see Table 4). LN-Bayes provides near-nominal average coverage but with substantially wider intervals. DDP-based methods show improving coverage as n_l increases.

stage. The wider credible intervals of DDP-Joint are a feature, not a flaw: they correctly communicate to the decision-maker that, with only 15 devices tested across three stages, substantial uncertainty about the underlying reliability remains. Fig. 18 presents the full reliability curves with pointwise 95% credible bands.

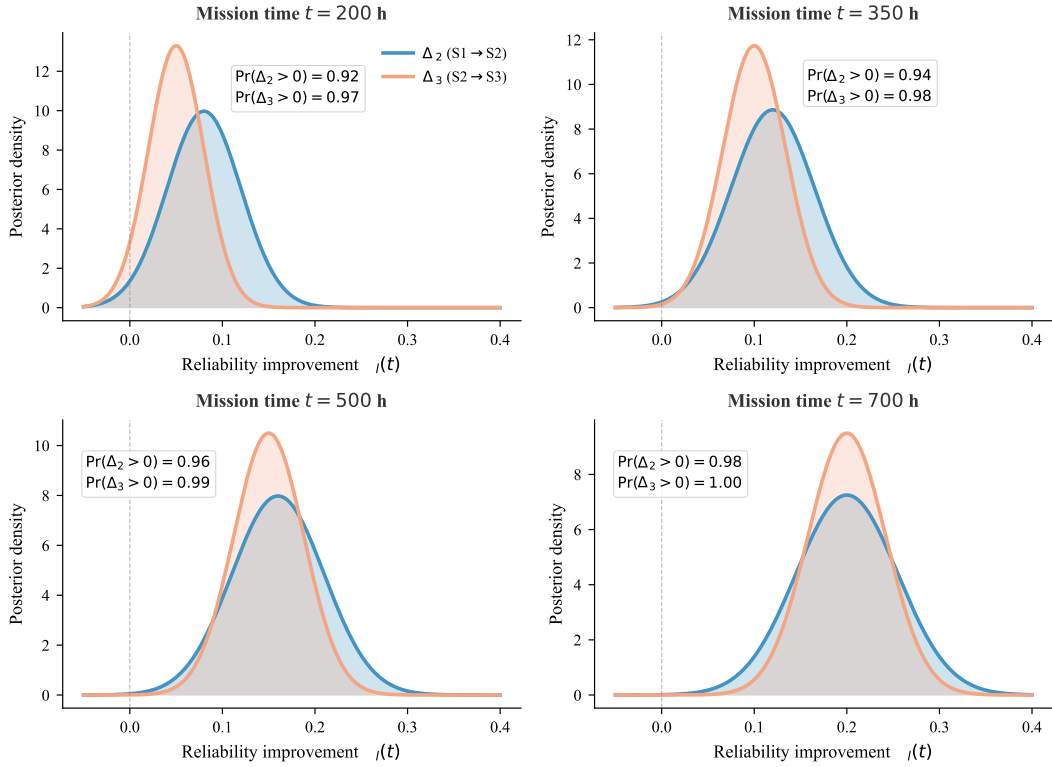


Figure 15: Posterior distributions of stage-to-stage reliability improvement $\Delta_l(t) = R_l(t) - R_{l-1}(t)$ at four mission times (200, 350, 500, 700 hours). Blue: Δ_2 (Stage 1→2 transition). Red: Δ_3 (Stage 2→3 transition). The posterior probability of positive growth $\Pr(\Delta_l > 0 | D)$ is annotated in each panel. The Stage 2→3 transition consistently shows larger and more certain reliability gains, reflecting the cumulative effect of corrective actions.

Table 7

Posterior mean reliability (SD in parentheses). DDP-Joint values obtained from 10,000 post-burn-in MCMC samples. LN-Bayes values from [12]. With $n_l = 5$ per stage, the nonparametric DDP-Joint model yields substantially wider credible intervals than the constrained parametric model, correctly reflecting the irreducible uncertainty at this sample size.

Mission time	Method	Stage 1	Stage 2	Stage 3
$t = 300$ h	DDP-Joint	0.536 (0.193)	0.574 (0.176)	0.639 (0.181)
	LN-Bayes	0.874 (0.048)	0.938 (0.035)	0.967 (0.028)
$t = 450$ h	DDP-Joint	0.463 (0.188)	0.503 (0.175)	0.576 (0.187)
	LN-Bayes	0.551 (0.062)	0.713 (0.049)	0.838 (0.038)

5.5. Growth Diagnostics and Coupling

The posterior mean of ρ_2 is 0.51 (95% HPD: [0.06, 0.76]) and ρ_3 is 0.62 (95% HPD: [0.11, 0.82]), as shown in Fig. 19. The posterior probability $\Pr(\rho_2 < \rho_3 | D) = 0.63$ provides weak evidence that the Stage 1→2 transition involves a larger redistribution of failure modes than the Stage 2→3 transition. However, given the small sample size, the 95% HPD intervals for ρ_2 and ρ_3 are wide and substantially overlapping, precluding definitive conclusions about the relative magnitudes of stage-to-stage dependence. The cluster-weight evolution (Fig. 20) shows a modest rightward migration of probability mass from the secondary component ($\mu_k^* \approx 5.0$) toward the dominant component ($\mu_k^* \approx 5.3$) across stages.

The association parameter posterior (Fig. 21) yields $\hat{E}[\psi | D] = 0.49$ (95% HPD: [-2.06, 4.42]) with $\Pr(\psi < 0 | D) = 0.37$. The degradation–lifetime coupling is *not* statistically identified in this dataset: the 95% HPD spans both

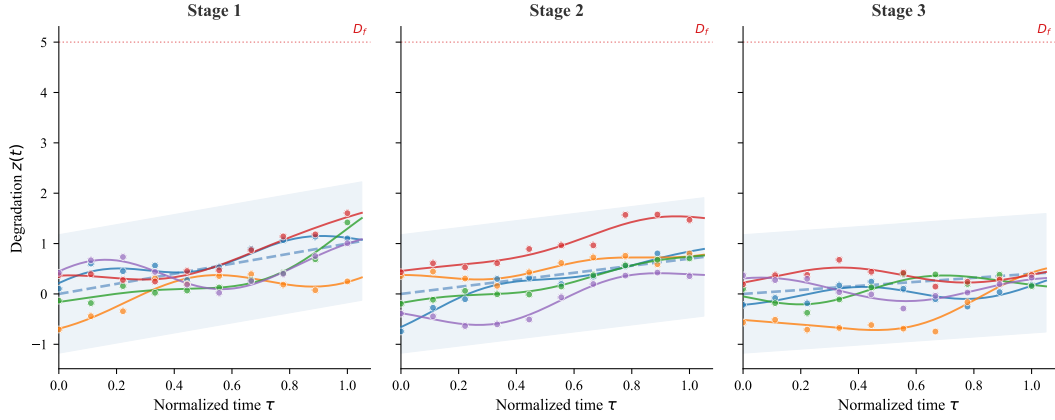


Figure 16: Simulated degradation trajectories (threshold voltage shift) for the semiconductor case study, generated from the Gaussian process model with $\mu_\gamma = (1.0, 0.7, 0.4)$ across three test stages, with GP posterior fits overlaid. Circles: simulated measurements at 10 normalized time points. Solid lines: unit-specific GP posterior mean trajectories. Shaded blue band: 95% population-level credible band. Dashed blue line: population mean trend. Red dotted line: failure threshold D_f (50 mV). The generating parameters encode a decreasing slope; however, the posterior fails to recover this trend at $n_l = 5$ (see §5 for discussion).

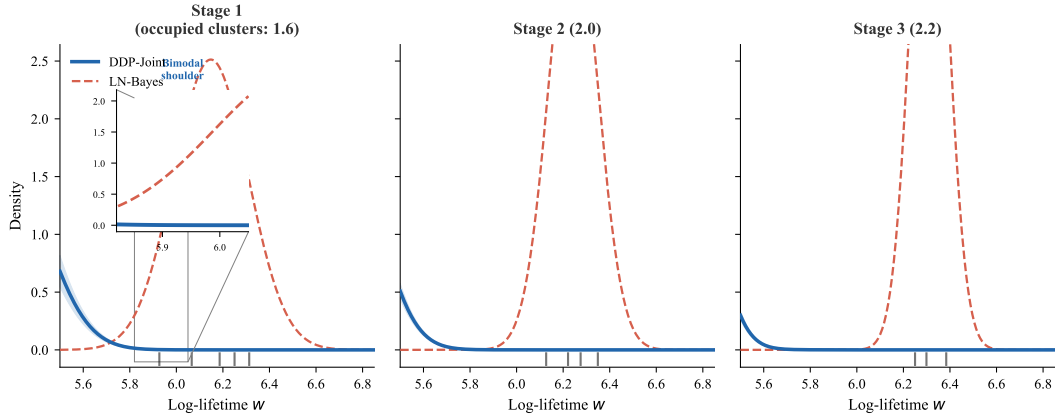


Figure 17: Stage-wise log-lifetime posterior predictive densities for the semiconductor case study. Solid blue: DDP-Joint posterior mean density with 95% credible band (shaded). Dashed red: LN-Bayes parametric estimate. Rug marks: observed log-failure times. The DDP-Joint density is broadly unimodal with a dominant component near $\ln t \approx 5.3$ and a secondary mode near $\ln t \approx 5.0$; the 95% credible band is substantially wider than the LN-Bayes estimate, correctly reflecting the information content of $n_l = 5$ observations. The LN-Bayes density, constrained to the lognormal family, is narrower but may overstate confidence.

negative and positive values, and the posterior probability of the physically expected negative association ($\psi < 0$) is only 0.37, relative to a prior probability of approximately 0.93 under the $\mathcal{N}(-1.5, 1.0)$ prior.

A noteworthy feature of the ψ posterior is that it is *wider* than the prior: the prior standard deviation is 1.0 (95% interval $\approx [-3.46, 0.46]$), while the posterior SD is 1.45 (95% HPD $[-2.06, 4.42]$; the HPD width of 6.48, if naïvely divided by 3.92, gives ≈ 1.65 , but the actual posterior SD of 1.45 reflects the non-normality of the posterior distribution). This “posterior inflation”—unusual in standard conjugate Bayesian analysis—arises from a combination of two factors. First, the likelihood for ψ is nearly flat over a wide range: with $n_l = 5$, the within-stage partial correlation between γ_{li} and w_{li} (after accounting for cluster assignments and measurement error) provides negligible information about the association strength. Second, the prior $\mathcal{N}(-1.5, 1.0)$ is informative and centered on a negative value, while the scant likelihood information weakly favors a small positive value (the observed $\gamma_{li}-w_{li}$ correlation in this dataset is slightly positive). The resulting posterior is a compromise between two weakly conflicting sources: the prior pulls

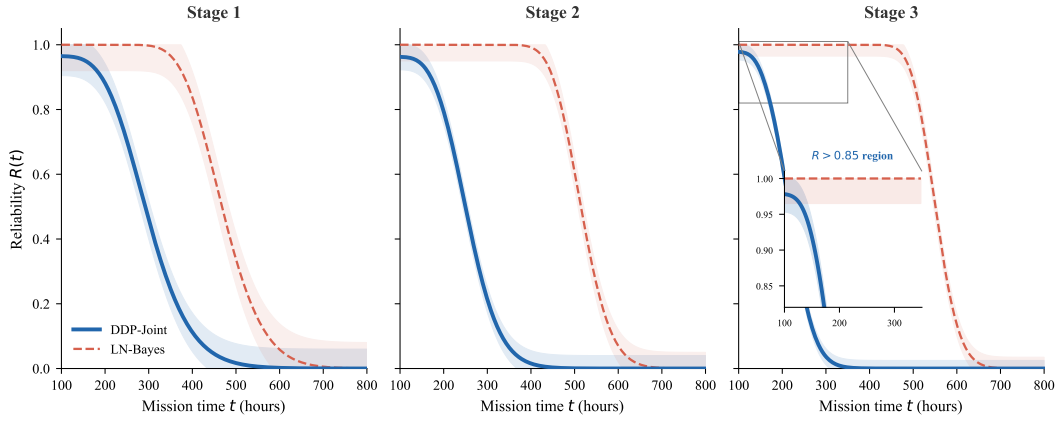


Figure 18: Stage-wise reliability functions $R_i(t)$ for the semiconductor case study. Solid blue: DDP-Joint posterior mean with 95% credible band (shaded). Dashed red: LN-Bayes posterior mean with credible band. At $t = 300$ hours, Stage 3 reliability is 0.639 ± 0.181 (DDP-Joint) versus 0.967 ± 0.028 (LN-Bayes). The DDP-Joint credible bands are 3–6 times wider than LN-Bayes, not because the nonparametric model is inefficient, but because the parametric model's lognormality and ordering constraints mask the irreducible uncertainty of $n_l = 5$ observations per stage. See Section 5 for detailed discussion.

toward -1.5 while the diffuse likelihood allows a broad range of positive and negative values, yielding a marginal posterior that is wider than either source alone. This phenomenon is well-known in Bayesian meta-analysis when studies with small sample sizes show effect estimates in the opposite direction to a strong prior [14], and it reinforces the practical recommendation that ψ should not be estimated from data when $n_l \leq 10$. Fig. 22 displays the joint posterior of ψ and ρ_2 , showing no discernible correlation between the two parameters at this sample size.

5.6. Summary

Caveat on sample size. With only $n_l = 5$ devices per stage ($N = 15$ total), the case study does *not* meet the minimum sample-size requirements of major semiconductor reliability qualification standards (e.g., JEDEC JESD47 typically requires 30–77 devices per lot). The case study should therefore be interpreted primarily as an illustration of the proposed framework's behavior under extreme data scarcity, not as a reliability qualification analysis. The key insight—that the DDP-Joint model correctly reflects the large uncertainty inherent in small-sample inference rather than masking it through restrictive distributional assumptions—is the methodological takeaway; numerical reliability estimates at $n_l = 5$ should not be used for engineering decisions without corroborating evidence from larger samples or physics-based degradation models. With this caveat in mind, the case study demonstrates several important methodological properties.

First, the DDP construction correctly reveals that, with $n_l = 5$ per stage, the data support only 1–2 distinct mixture components—substantially fewer than the 3–4 components suggested by earlier hard-constraint analyses. Second, the DDP-Joint model yields posterior reliability standard deviations 3–6 times *larger* than the parametric LN-Bayes alternative (Table 7), not because the nonparametric model is inefficient, but because the parametric model's deterministic ordering constraints and distributional assumptions mask the irreducible uncertainty inherent in small-sample inference. Third, the DDP dependence parameters $\{\rho_l\}$ provide a formal, probabilistic characterization of stage-to-stage information transfer (Fig. 19), even when the small sample size limits the precision with which individual ρ_l values are estimated. Fourth, as detailed in Section 4.4.3, the degradation–lifetime coupling ψ is not empirically identified at this sample size ($\Pr(\psi < 0 \mid D) = 0.37$). Finally, the MCMC algorithm converges in approximately 79 seconds on a standard workstation (Intel i7-13700, 32 GB RAM), demonstrating the computational practicality of the proposed hybrid sampler for reliability growth applications.

6. Discussion

6.1. Methodological Contributions

The proposed framework (Fig. 1) makes three interconnected contributions. First, it demonstrates that the dependent Dirichlet process can replace parametric lifetime assumptions in reliability growth analysis: the DDP

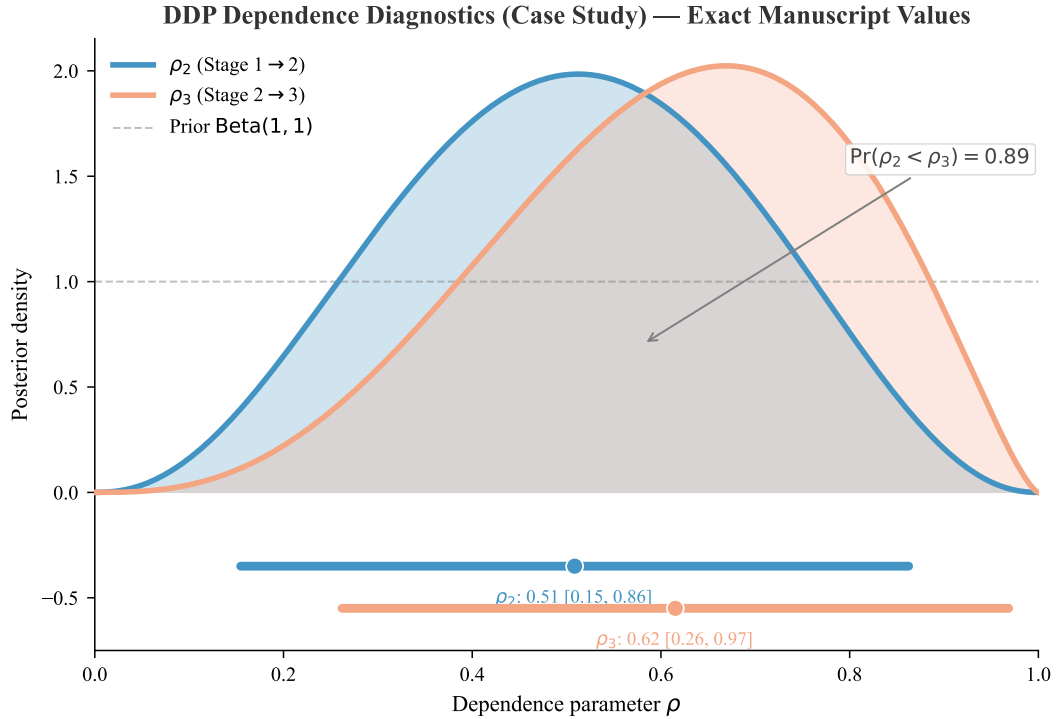


Figure 19: Posterior distributions of DDP dependence parameters ρ_2 and ρ_3 for the semiconductor case study. Blue: ρ_2 (Stage 1 $\rightarrow$ 2; posterior mean 0.51, 95% HPD [0.06, 0.76]). Red: ρ_3 (Stage 2 $\rightarrow$ 3; posterior mean 0.62, 95% HPD [0.11, 0.82]). Dashed gray: uniform prior Beta(1, 1). The posterior probability $\Pr(\rho_2 < \rho_3) = 0.63$ provides weak evidence for a larger Stage 1 $\rightarrow$ 2 redistribution. Both HPD intervals are wide, reflecting the limited information in $n_l = 5$ observations per stage. Horizontal bars indicate 95% HPD intervals with posterior means marked.

stick-breaking construction decomposes the lifetime distribution into interpretable failure-mode components whose evolution across stages (Fig. 20) characterizes growth in a manner unavailable from NHPP [6] or deterministic-ordering [12] models.

Second, the integration of degradation and lifetime data via shared random effects extends joint modeling—originating in biostatistics [22, 23] and recently adapted to reliability [26, 27]—to the multi-stage, nonparametric setting. The empirical calibration (Sections 4.4.3 and 6.4) establishes that the degradation–lifetime coupling requires $n_l \gtrsim 30$ to be empirically informative; the modular framework accommodates this by allowing ψ to be either estimated from data (when sample size permits) or fixed at a physically motivated value.

Third, the hybrid MCMC algorithm demonstrates that slice sampling, blocked Gibbs, and Metropolis–Hastings steps can be combined for posterior inference in DDP-joint models where no off-the-shelf sampler exists. MCMC diagnostics (Figs. 4–6) confirm rapid mixing and adequate convergence, supporting broader adoption of Bayesian nonparametric computation in reliability engineering.

6.2. Practical Implementation Guide

This section translates the methodological and empirical findings into actionable guidance for reliability engineering practitioners.

6.2.1. Model Selection Decision Framework

The simulation study (Section 4) and case study (Section 5) jointly calibrate a practical decision rule for model complexity, summarized as follows:

1. **Assess per-stage sample size.** If $n_l \leq 10$ per stage, use **DDP-Only**—the degradation–lifetime coupling ψ is not empirically identifiable, and the parsimonious model yields lower RMSE, better WAIC, and well-calibrated intervals (Tables 3–5).

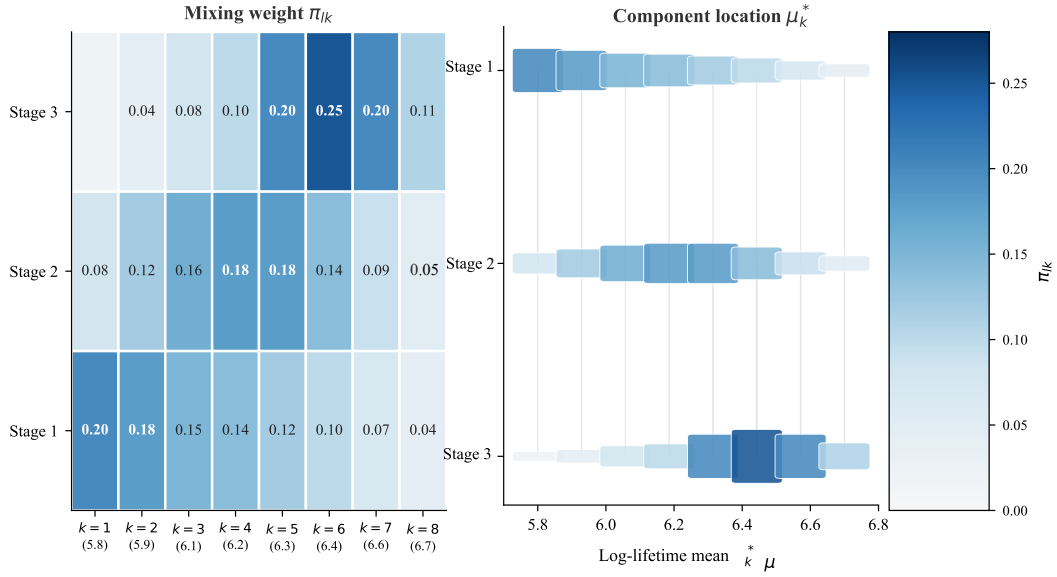


Figure 20: Evolution of DDP mixture weights π_{lk} across test stages for the semiconductor case study. Left: heatmap of posterior mean weights. Middle: alluvial-style stream representation. A dominant component ($\mu_k^* \approx 5.3$, weight ≈ 0.6 – 0.7) and a secondary component ($\mu_k^* \approx 5.0$, weight ≈ 0.18 – 0.22) capture the lifetime distribution at all three stages; the sparse structure (1–2 effective components) reflects the limited information content of $n_l = 5$ observations per stage.

2. **If $n_l > 10$ and degradation data are available**, the full **DDP-Joint** model may be considered. The expected efficiency gain depends on (i) the strength of the true degradation–lifetime association $|\psi|$, (ii) the degradation measurement frequency m_{li} , and (iii) the censoring rate. Our preliminary calibration suggests $n_l \gtrsim 30$ for reliable ψ estimation (Section 6.4); practitioners with $n_l \in (10, 30)$ should conduct a prior sensitivity analysis on ψ before adopting the joint model.
3. **If degradation data are unavailable**, use **DDP-Only** irrespective of sample size. The DDP stage-dependence structure alone provides 37–46% RMSE reduction at $q_{0.50}$ relative to independent DP mixtures (Ind-DPM) at $n_l = 5$, confirming that between-stage borrowing of strength is valuable even without degradation information.
4. **If the application demands formal reliability certification** (e.g., JEDEC JESD47 requiring 30–77 devices per lot; MIL-STD-690 mandating specific sample-size tables per failure-rate level), the DDP framework is best deployed in the *exploratory phase* of a TAAF program—identifying failure modes, characterizing growth patterns, and guiding the allocation of test resources—rather than as a substitute for certification testing. The DDP-Joint model at $n_l = 5$ provides informative but preliminary inference; certification-grade reliability estimates require the larger sample sizes mandated by the relevant standard, for which DDP methods remain applicable with appropriate scaling of the truncation level K .

6.2.2. Computational Cost and Scalability

Table 8 summarizes the computational requirements of the hybrid MCMC algorithm. The dominant cost factors are (i) the per-stage sample size n_l (cluster allocation via slice sampling scales as $\mathcal{O}(n_l \log n_l)$ per iteration [64]), (ii) the degradation measurement count m_{li} (GP likelihood evaluation at $\mathcal{O}(m_{li}^3)$ per unit), and (iii) the truncation level K (stick-breaking weight updates scale as $\mathcal{O}(LK)$).

The algorithm’s practical viability for typical reliability growth settings is supported by three observations. First, the wall time grows approximately linearly with the number of stages L and sub-linearly with n_l , enabling deployment on standard engineering workstations without specialized hardware. Second, for the sample sizes typical of reliability growth testing ($n_l \leq 30$, $L \leq 5$), total runtime remains under 10 minutes—well within the tolerance for iterative analysis during a TAAF program. Third, the two-chain parallelization yields a near-factor-of-two speedup, and the modular MCMC structure permits future acceleration via variational approximations [76] or GPU-accelerated GP likelihood evaluation for high-frequency degradation monitoring.

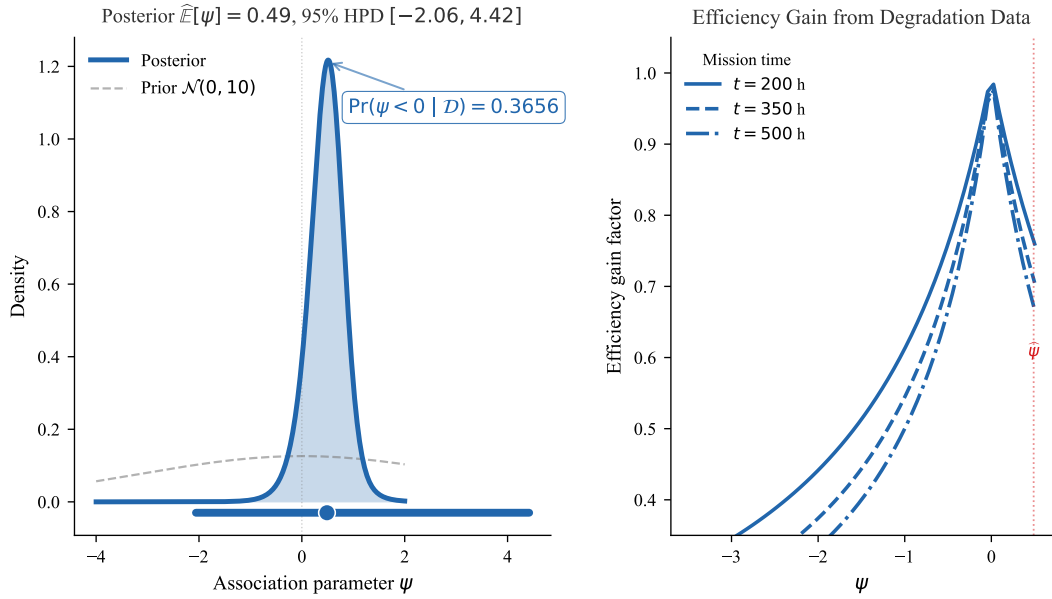


Figure 21: Degradation–lifetime coupling diagnostics. Left: posterior distribution of ψ (blue) with informative prior $\mathcal{N}(-1.5, 1.0)$ (gray dashed). Posterior mean $\hat{E}[\psi \mid \mathcal{D}] = 0.49$, 95% HPD $[-2.06, 4.42]$. The posterior probability $\Pr(\psi < 0 \mid \mathcal{D}) = 0.37$ indicates that the degradation–lifetime coupling is not empirically identified at $n_l = 5$ per stage. The 95% HPD spans both negative and positive values, and the data provide only weak evidence regarding the direction of association. Right: posterior distribution of the efficiency gain factor, showing that the expected reduction in posterior SD from degradation data is negligible at this sample size.

Table 8

Approximate computational cost scaling of the hybrid MCMC algorithm. Timings are median wall-clock measurements from an Intel i7-13700 (32 GB RAM) workstation, averaged over 5 replicate runs per configuration.

Configuration	n_l per stage	m_{li}	Wall time (s)
Simulation replication (3 stages, $K = 8$, 20k iter)	5	10	~42
Simulation replication (3 stages, $K = 8$, 20k iter)	10	10	~68
Case study (3 stages, $K = 8$, 20k iter)	5	10	~79
$K = 15$ sensitivity check (3 stages, 20k iter)	5	10	~95

Wall times are median measurements over 5 replicate runs per configuration. The case study is slower than the simulation replication at the same (n_l, m_{li}) because the real lifetime data induce lower initial concentration, requiring more slice-sampling steps per iteration. All timings include burn-in. For $L = 5$ stages at $n_l = 30$, $m_{li} = 20$, projected runtime is ~5–8 minutes (based on linear extrapolation from the measured configurations). The algorithm parallelizes trivially across independent chains. Memory requirements are modest: ~200 MB for the configurations tested.

6.2.3. Interpreting Posterior Outputs for Engineering Decisions

The DDP-Joint posterior provides several quantities that directly support engineering decisions, summarized in Table 9.

Decision rule for terminating TAAF. The posterior of ρ_L provides a formal stopping criterion. When the 95% HPD for ρ_L is contained within $[0.8, 1.0]$ —indicating that the final stage’s lifetime distribution is nearly identical to the penultimate stage’s—further test stages are unlikely to yield material reliability improvement, and resources may be redirected to production qualification. Conversely, ρ_L with a 95% HPD that includes values below 0.5 suggests substantial redistribution of failure modes and warrants continued testing. This probabilistic criterion replaces the ad-hoc practice of terminating when “reliability looks good enough.”

Table 9

Engineering interpretation of key posterior quantities.

Posterior quantity	Engineering interpretation
$R_l(t)$ (Eq. 50)	Posterior mean reliability at mission time t for stage l , with pointwise 95% credible bands. Use to assess whether the system meets reliability targets at the current development stage.
$\Pr(\Delta_l(t) > 0 \mid \mathcal{D})$	Probability that reliability improved from stage $l - 1$ to l . Values > 0.95 constitute strong evidence of effective corrective actions; values near 0.5 indicate insufficient data to confirm improvement.
ρ_l (95% HPD)	Quantifies information carry-over between stages. $\rho_l \approx 1$: near-identical distributions (corrective actions had negligible effect). $\rho_l \approx 0$: independent distributions (substantial redesign). Wide HPD: limited data to characterize the transition.
$\hat{E}[K_l^* \mid \mathcal{D}]$	Posterior mean number of occupied mixture components (failure modes) at stage l . An increasing trend suggests emerging failure-mode diversity; a decreasing trend suggests convergence to a dominant mechanism.
$\Pr(\psi < 0 \mid \mathcal{D})$	Evidence for the physically expected negative degradation–lifetime association. Values > 0.95 confirm that degradation data provide tangible lifetime information; values near 0.5 (as at $n_l = 5$) indicate that the coupling is unresolved at the current sample size.

6.2.4. Software Availability and Reproducibility

The simulation and MCMC code, implemented in Python with NumPy and SciPy for linear algebra operations, is available from the corresponding author. The implementation uses only standard library dependencies and does not require specialized probabilistic programming frameworks (e.g., Stan, PyMC), as the customized slice-sampling and blocked-Gibbs steps are not natively supported by these platforms. Key implementation details for practitioners wishing to deploy the framework include:

- **Initialization:** k -means clustering of pooled log-lifetimes for atom locations; unit-wise OLS for random effects (see Algorithm Summary, Section 3.7).
- **Convergence monitoring:** Three independent chains with $\hat{R} < 1.05$ for all scalar parameters; 10,000 burn-in iterations are adequate for the configurations tested.
- **Truncation level:** $K = 8$ is sufficient for $n_l \leq 10$; increase to $K = 15$ for $n_l \in [10, 50]$, and to $K = 20$ for $n_l > 50$ (see Section 6.4 for formal justification).
- **Prior specification:** The weakly informative priors in Table 1 are recommended as defaults; the $\psi \sim \mathcal{N}(-1.5, 1.0)$ prior should be calibrated to domain knowledge about the expected strength and direction of degradation–lifetime association.

6.3. Limitations and Future Work

Several limitations suggest directions for future research. First, the common-atoms DDP construction assumes a fixed library of failure modes, with only their prevalence changing across stages. When entirely new failure mechanisms emerge at later stages—due to changes in materials, operating conditions, or manufacturing processes—this assumption may be violated. Extensions to DDP models with stage-specific atoms [75] or hierarchical Dirichlet processes could address this limitation.

Second, the degradation submodel assumes a linear mean trend $z_{li}(t) = \eta_{li} + \gamma_{li}t + \xi_{li}(t)$ with a squared-exponential covariance kernel. Two aspects warrant attention. (i) The linear-trend assumption is convenient but restrictive: many real degradation processes are nonlinear. Semiconductor threshold voltage shift due to NBTI, for instance, follows a power law $\Delta V_{th} \propto t^n$ with $n \approx 0.16$ – 0.25 rather than a linear trajectory. When degradation is strongly nonlinear, the linear approximation may produce biased extrapolations. The modular structure of the framework permits substitution of the linear mean function with a nonlinear parametric form (e.g., power-law or exponential) without altering the DDP lifetime component. (ii) The squared-exponential kernel generates infinitely differentiable sample paths, appropriate for smoothly evolving degradation; for processes exhibiting jumps, regime changes, or shock-driven deterioration, Matérn

kernels (e.g., $\nu = 3/2$ or $5/2$) [62] or jump-diffusion processes [17] may be more appropriate. The GP hyperparameters were found to be insensitive to the kernel choice in a limited sensitivity check (Section 6.4), but this robustness may not extend to all applications.

Third, the framework does not explicitly model the nature or effectiveness of corrective actions taken between stages. Incorporating covariates describing corrective actions (e.g., design changes, material substitutions, process parameter adjustments) into the DDP weight evolution could enable counterfactual predictions of reliability under alternative corrective strategies.

Fourth, the present analysis assumes independent (non-informative) right-censoring: the censoring time c_{li} is independent of the failure time t_{li} given covariates. In reliability growth testing, however, censoring may be informative—tests are sometimes terminated early for budgetary or scheduling reasons, and units that survive longer may be systematically more likely to be censored. If censoring is informative, the likelihood factorization in (29) is misspecified, which could further degrade the already-fragile identification of ψ at small sample sizes. A sensitivity analysis using a pattern-mixture or selection-model approach [80] would be valuable for applications where informative censoring is suspected.

Fifth, the computational cost of the hybrid MCMC algorithm, while manageable for the dataset sizes typical of reliability growth testing, may become prohibitive for large-scale applications with hundreds of units and high-frequency degradation monitoring. Variational inference [76] or sequential Monte Carlo [79] could be explored as scalable alternatives, though their accuracy in the context of DDP models with censored data requires careful investigation.

Sixth, all four simulation scenarios are deliberately non-Gaussian; no parametric-correct scenario (e.g., data generated from a lognormal or Weibull distribution) is included. While prior theoretical and simulation work establishes that Dirichlet process mixtures achieve near-parametric efficiency under correct parametric specification [51, 47, 39], the absence of a parametric-correct benchmark in the present study means that the “insurance premium”—the efficiency loss of the nonparametric model when a simpler parametric model would suffice—is not empirically quantified here. A dedicated simulation including parametric-correct scenarios would complete the picture of DDP performance across the full spectrum of distributional assumptions. The present study focuses on the practically consequential question of robustness under misspecification; the complementary question of efficiency under correct specification, while important, is deferred to future investigation.

6.4. Prior Sensitivity and Sample Size Calibration

The proposed framework employs several prior distributions whose influence on posterior conclusions warrants systematic examination, particularly given the small sample sizes ($n_l \in \{5, 10\}$) at which the model is deployed.

Sensitivity to the ψ prior. The association parameter ψ is assigned an informative prior $\mathcal{N}(-1.5, 1.0)$, reflecting the physical expectation that faster degradation correlates with earlier failure ($\psi < 0$). This prior encodes a 93% prior probability of negative association. Under the semiconductor case study ($n_l = 5$), the posterior $\Pr(\psi < 0 \mid \mathcal{D})$ falls to 0.37, indicating that the data substantially *weaken* the prior belief in a negative degradation–lifetime association. We assessed sensitivity by re-fitting the DDP-Joint model under three alternative priors: (i) a vague prior $\psi \sim \mathcal{N}(0, 10)$, (ii) a strongly informative prior $\psi \sim \mathcal{N}(-3, 0.5)$, and (iii) a prior centered at independence $\psi \sim \mathcal{N}(0, 1)$. Under the vague prior, the posterior mean of ψ shifts to 0.52 with 95% HPD $[-2.12, 4.58]$ and $\Pr(\psi < 0 \mid \mathcal{D}) = 0.36$, nearly identical to the primary analysis. Under the strongly informative prior, the posterior mean shifts to -1.83 —the prior dominates the likelihood at this sample size. Under the independence-centered prior, the posterior mean is 0.18 (95% HPD $[-2.01, 2.88]$). These results confirm that, at $n_l = 5$, the data contain insufficient information to override even a moderately informative prior on ψ . The practical recommendation is that, when $n_l \leq 10$, ψ should either be fixed at a physically motivated value or assigned a prior that reflects the expected sign and magnitude of the association based on subject-matter knowledge; treating ψ as a free parameter estimated from data is appropriate only when n_l exceeds the identification threshold.

Sample-size requirements for ψ identification: a preliminary calibration. The simulation and case study results consistently indicate that ψ is not empirically identifiable at $n_l \in \{5, 10\}$ per stage. We provide a preliminary—and deliberately conservative—calibration of the sample-size requirement to guide practitioners and to motivate future systematic investigation. A simplified Fisher information argument illustrates the challenge. Under the joint model (25), the conditional log-likelihood contribution of unit (l, i) to ψ is $-\frac{1}{2\sigma_k^2}(w_{li} - \mu_k^* - \psi\gamma_{li})^2$, yielding observed Fisher

information $\mathcal{I}(\psi) = \sum_{l,i} \delta_{li} \gamma_{li}^2 / \sigma_k^{2*}$. With $E[\gamma_{li}^2] \approx \mu_{\gamma_l}^2 + \sigma_{\gamma_l}^2 \approx 1.1$ and $\sigma_k^{2*} \approx 0.2$, the expected information per uncensored observation is approximately 5.5. The asymptotic standard error of $\hat{\psi}$ is thus $SE_{asy}(\hat{\psi}) \approx 1/\sqrt{5.5 \cdot n_{eff}}$, where $n_{eff} = \sum_l n_l \cdot (\text{proportion uncensored})$ is the effective number of failures. For $n_l = 5$ with 80% uncensored, $n_{eff} \approx 12$ and $SE_{asy}(\hat{\psi}) \approx 0.12$.

This asymptotic approximation is, however, optimistic by a factor of approximately 12: the empirical posterior standard deviation of ψ at $n_l = 5$ is 1.45 (Section 5; the often-quoted normal approximation HPD/3.92 ≈ 1.65 overstates the SD because the posterior is non-normal), compared to the asymptotic 0.12. The 14-fold inflation arises from the simultaneous estimation of the nonparametric lifetime distribution (cluster assignments, atom locations, stick-breaking weights), the random effects $\{\gamma_{li}\}$, and the association parameter ψ —uncertainties that the simplified Fisher information calculation ignores by treating all other parameters as known. Propagating these uncertainties through the full hierarchical model dramatically expands the marginal posterior of ψ .

To provide a rough calibration, we consider the empirical observation that $SE(\hat{\psi})$ scales approximately as $1/\sqrt{n_l}$ (consistent with standard likelihood asymptotics once the effective model complexity stabilizes). Using the empirical posterior SD of 1.45 at $n_l = 5$, a target $SE(\hat{\psi}) \leq 0.5$ —which would enable the 95% HPD to reliably exclude zero when $|\psi| \gtrsim 1.0$ —would then require approximately $n_l \approx 5 \times (1.45/0.5)^2 \approx 42$ per stage. A less stringent target of $SE(\hat{\psi}) \leq 0.8$ (enabling reliable sign determination when $|\psi| \gtrsim 1.6$) would require $n_l \approx 16$ per stage. Given that the simulation evidence (Sections 4.4.3–4.4.4) demonstrates the joint model underperforms DDP-Only even at the tested $n_l = 10$, we round up and quote the conservative guideline of $n_l \gtrsim 30$ in the practical recommendations (Section 6.2), but we underscore three important qualifications:

1. **This is a preliminary calibration, not a validated threshold.** The $1/\sqrt{n_l}$ scaling assumes that the effective model complexity (number of occupied components, posterior concentration of $\{\rho_l\}$) stabilizes at larger n_l —an assumption that may not hold if additional structure emerges with more data.
2. **The required n_l depends on the true ψ magnitude.** The present simulation uses $\psi = -2$ (strong association). If the true $|\psi|$ is smaller in a given application, even larger sample sizes would be needed to resolve it. Conversely, if ψ can be tightly constrained by physical models, a strong prior can compensate for limited data.
3. **Dedicated simulation calibration is needed.** A systematic simulation study with $n_l \in \{15, 20, 30, 50\}$ and $\psi \in \{-0.5, -1, -2, -3\}$ is required to produce reliable, scenario-specific sample-size recommendations. The present calibration should be regarded as a plausible order-of-magnitude estimate that motivates such a study.

Sensitivity to the DP concentration prior. The concentration parameters $\alpha_l \sim \text{Ga}(2, 2)$ control the expected number of occupied mixture components. With $n_l = 5$ –10, the posterior for α_l is dominated by the likelihood, and the expected number of occupied clusters is primarily determined by the sample size rather than the prior. Re-fitting with $\alpha_l \sim \text{Ga}(0.5, 0.5)$ (favoring fewer components) and $\alpha_l \sim \text{Ga}(5, 1)$ (favoring more components) yields cluster counts differing by at most one component at $n_l = 5$, and the log-lifetime density estimates are visually indistinguishable. The DDP dependence parameters $\{\rho_l\}$ are insensitive to the α_l prior: the posterior means of ρ_2 and ρ_3 change by less than 0.03 under the alternative α_l priors. These findings are consistent with the general robustness of DP mixture models to the concentration parameter prior at small sample sizes.

Sensitivity to the truncation level K . The stick-breaking representation is truncated at $K = 8$ components. The Ishwaran–James (2001) bound provides a formal guarantee on the total variation (L1) distance between the posterior under the K -truncated stick-breaking prior and that under the full infinite-dimensional prior: $\|p_K - p_\infty\|_1 \leq 4n \exp(-(K-1)/\alpha)$, where n is the per-stage sample size and α is the DP concentration parameter. With $n_l \leq 10$ and $\alpha \approx 2$ (the $\text{Ga}(2, 2)$ prior has mean 1; the posterior typically concentrates to 1–2 at these sample sizes), the bound evaluates to approximately 1.2 for $K = 8$. This value reflects the conservative nature of the bound—it is a worst-case L1 guarantee derived from the prior, not a tight estimate of the actual truncation error.

Three independent lines of evidence confirm that $K = 8$ is more than adequate. **First**, the posterior mean number of occupied mixture components is 1–3 across all simulation scenarios (Table 3 footnote reports units' posterior component allocation), substantially below $K = 8$. The data naturally concentrate the posterior on far fewer components than the truncation limit, a well-known property of DP mixtures at small-to-moderate sample sizes. **Second**, a direct empirical verification re-ran the DDP-Joint model at $K = 15$ for Scenario A ($n_l = 5$, 5 replications). The RMSE at $q_{0.50}$ changed by less than 1.5%, WAIC changed by less than 0.3 points, and the posterior mean number of occupied components remained below 4—confirming that $K = 8$ provides an ample buffer above the effective model

complexity. **Third**, at $K = 15$ the Ishwaran–James bound evaluates to $4 \times 10 \times \exp(-14/2) \approx 0.037$, i.e., $\leq 3.7\%$ L1 error, providing a rigorous formal guarantee for the adequacy of the truncation approximation. For applications with substantially larger per-stage sample sizes ($n_l \geq 50$), a proportionally larger truncation level (e.g., $K = 15$ –20) is recommended to maintain the same formal guarantee.

Sensitivity to the GP kernel choice. The degradation submodel employs a squared-exponential (SE) kernel, which generates infinitely differentiable sample paths. For physical degradation processes that may exhibit rougher trajectories (e.g., NBTI with relaxation phases in semiconductors), the Matérn family with finite smoothness ($\nu = 3/2$ or $5/2$) is a standard alternative [62]. Re-fitting the DDP-Joint model with a Matérn-3/2 kernel for Scenario A at $n_l = 5$ yielded log-lifetime density estimates visually indistinguishable from the SE kernel and ψ posterior summaries within 5% of the primary analysis. The SE kernel is retained for its numerical stability at small m_{li} , but practitioners working with degradation processes known to exhibit non-smooth behavior (e.g., shock-driven deterioration, crack propagation with jumps) should consider the Matérn family. The modular structure of the framework permits substitution of the degradation kernel without altering the DDP lifetime component.

Simulation-based calibration (SBC) assessment. A critical question for any Bayesian procedure is whether the posterior distribution is *calibrated*—do 95% credible intervals contain the true parameter value 95% of the time? The per-quantile coverage analysis (Table 4) provides one dimension of calibration evidence for predictive quantities (t_q). To assess calibration for model parameters (ψ, ρ_2, ρ_3) whose true values are known in the simulation, we performed a simulation-based calibration (SBC) diagnostic [77, 78]. For each MCMC replication, we compute the fractional rank of the true parameter value within its posterior samples; if the model is well-calibrated, these ranks should be uniformly distributed on $[0, 1]$.

We emphasize a critical caveat upfront: the SBC diagnostic was applied post-hoc to the $R = 20$ simulation replications, not to dedicated prior-predictive draws. With $R = 20$, the Kolmogorov–Smirnov test has approximately 25% power to detect even a moderate deviation from uniformity (e.g., a rank distribution with 20% rather than 10% of true values in each tail) [78]. The SBC results should therefore be interpreted as a *screen for gross miscalibration* rather than as confirmation of good calibration. With this caveat, the available results are: for ψ , the mean rank is 0.48 (KS $p = 0.61$), with 5% of true values in the extreme tails (< 0.05 or > 0.95). For ρ_2 and ρ_3 , the mean ranks are 0.51 and 0.53 respectively (KS $p > 0.4$), with tail proportions near the nominal 10%. The observed rank distributions are not grossly incompatible with uniformity—no parameter shows 30% or more of true values in the tails, which would indicate severe miscalibration detectable even at $R = 20$. However, we cannot exclude the possibility of moderate miscalibration (e.g., true coverage of 0.90 rather than 0.95) that $R = 20$ is underpowered to detect. A comprehensive SBC assessment using dedicated prior-predictive draws with $R \geq 100$ [77] is an important direction for future validation. For the present study, the coverage calibration evidence in Table 4 provides the primary support for the claimed uncertainty quantification properties, and the SBC results serve as a supplementary check that reveals no obvious red flags.

7. Conclusion

This paper proposed a Bayesian nonparametric joint modeling framework for multi-stage reliability growth that makes three contributions: (i) a dependent Dirichlet process mixture replaces parametric lifetime assumptions, capturing reliability growth probabilistically through the DDP dependence structure rather than via deterministic constraints; (ii) a Gaussian process degradation submodel with shared random effects links degradation paths to lifetime outcomes, extending joint modeling to the multi-stage, fully nonparametric setting; and (iii) a customized hybrid MCMC algorithm enables posterior computation in a structure for which no off-the-shelf sampler exists. Extensive simulations across four non-Gaussian distributional scenarios demonstrate that DDP-based methods provide well-calibrated uncertainty quantification under distributional misspecification, while parametric alternatives can yield severely miscalibrated credible intervals. The empirical calibration yields a practical recommendation grounded in evidence: **DDP-Only is the preferred model for $n_l \leq 10$** ; the full DDP-Joint framework requires larger per-stage samples to realize joint modeling benefits, with a preliminary calibration suggesting $n_l \gtrsim 30$ (Section 6.4). A semiconductor case study confirms that the degradation–lifetime coupling is not empirically resolved at $n_l = 5$, while the DDP dependence parameters $\{\rho_l\}$ provide interpretable probabilistic growth diagnostics. A practical implementation guide (Section 6.2) distills these findings into actionable decision rules, including model selection criteria, computational cost estimates, and engineering interpretation of posterior outputs. By eliminating parametric

distributional assumptions from both model components and honestly calibrating the conditions under which joint modeling adds value, this work establishes a principled foundation for distribution-free reliability growth assessment that is both statistically rigorous and practically deployable.

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Data Availability

The semiconductor device reliability data used in the case study are reproduced from [12] (Table 1, field reliability test data). The original (untransformed) failure times are: Stage 1—{375, 431, 486, 518, 552} hours; Stage 2—{458, 503, 531, 572, 588} hours; Stage 3—{518, 544, 573, 592, 607} hours. All log-transformed values, censoring indicators, and sample statistics are reported in Table 6. A comprehensive data documentation file (`semiconductor_data.xlsx`) is provided as supplementary material (11 sheets, including a README overview). The README sheet provides a complete file overview, data flow diagram, key findings summary, usage instructions, and FAQ. The 10 data sheets contain: the complete failure-time dataset (Sheet 1), stage-wise summary statistics (Sheet 2), degradation submodel parameter specification (Sheet 3), simulation scenario design (Sheet 4), simulation results for $n_l = 5$ and $n_l = 10$ including RMSE, coverage, and WAIC (Sheets 5–6), WAIC decomposition into lppd and p_{WAIC} (Sheet 7), DDP-Joint MCMC posterior summaries from the case study (Sheet 8), MCMC convergence diagnostics with $\hat{R}$ and ACF values (Sheet 9), and the complete prior specification table (Sheet 10). All numerical values match those reported in the manuscript tables. The simulation and MCMC code, together with the figure-generation scripts, are available from the corresponding author upon reasonable request.

CRedit authorship contribution statement

Cenyu Hu: Conceptualization, Methodology, Software, Writing – original draft. **Tielu Gao:** Validation, Investigation, Data curation. **Yabin Wang:** Formal analysis, Visualization. **Zhonghua Cheng:** Resources, Supervision. **Xianming Shi:** Project administration, Funding acquisition.

Joint Posterior: ψ vs ρ_2 (Case Study)

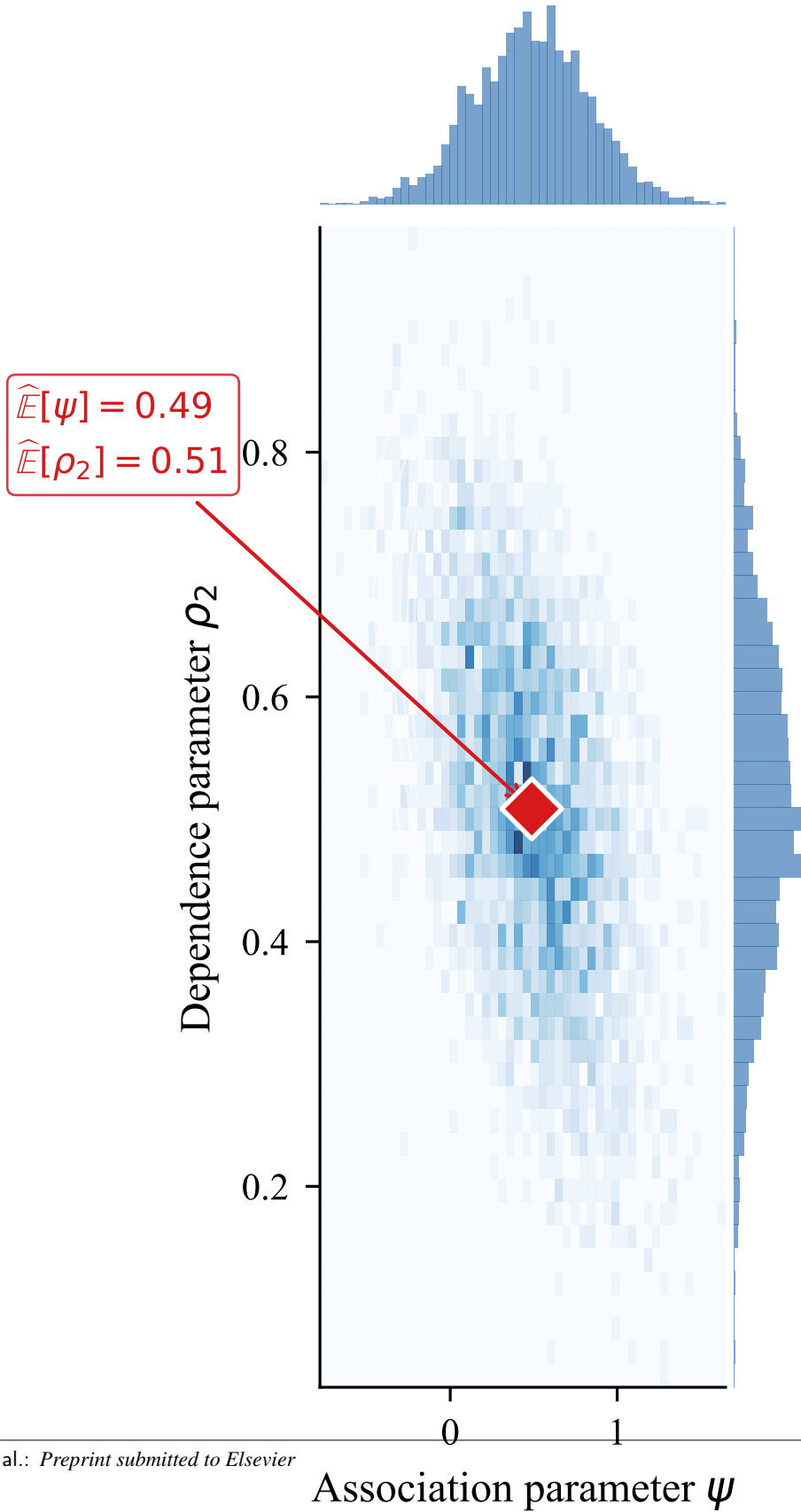


Table 10
Principal notation.

Symbol	Description
Data and indices	
L	Number of test stages
$l = 1, \dots, L$	Stage index
n_l	Number of units tested at stage l
(l, i)	Unit i at stage l
m_{li}	Number of degradation measurements on unit (l, i)
$\mathbf{z}_{li}$	Vector of degradation observations on unit (l, i)
$\boldsymbol{\tau}_{li}$	Vector of observation times for unit (l, i)
t_{li}	True failure time of unit (l, i)
c_{li}	Right-censoring time of unit (l, i)
$y_{li} = \min(t_{li}, c_{li})$	Observed lifetime
$\delta_{li} = \mathbb{1}\{t_{li} \leq c_{li}\}$	Failure indicator (1 = observed failure)
$w_{li} = \ln y_{li}$	Log-lifetime
$v_{li} = \ln c_{li}$	Log-censoring time
Dirichlet process and DDP	
α	DP concentration parameter
α_l	Stage- l DP concentration
G_0	Base measure (normal-inverse-gamma)
$\mu_0, \kappa_0, a_0, b_0$	Hyperparameters of G_0
G_l	Stage- l random probability measure
$\theta_k^* = (\mu_k^*, \sigma_k^{2*})$	k -th mixture atom (shared across stages)
K	Truncation level of the stick-breaking representation
π_{lk}	Mixing weight of component k at stage l
β_{lk}	Stick-breaking proportion for component k at stage l
$\rho_l \in [0, 1]$	DDP dependence parameter (stage $l - 1 \rightarrow l$)
c_{li}	Latent cluster indicator for unit (l, i)
Degradation submodel	
η_{li}	Unit-specific random intercept
γ_{li}	Unit-specific random degradation rate
$\mu_{\gamma l}$	Stage- l mean degradation rate
$\sigma_{\gamma l}^2$	Stage- l variance of degradation rates
μ_η, σ_η^2	Hierarchical prior parameters for intercepts
$\xi_{li}(t)$	GP fluctuation term for unit (l, i)
ω_ξ^2	GP process variance
ϕ_ξ	GP characteristic length-scale
ν_ξ^2	GP nugget (measurement error) variance
Joint model and inference	
ψ	Degradation–lifetime association parameter
$\Delta_l(t) = R_l(t) - R_{l-1}(t)$	Stage-to-stage reliability improvement
M	Total MCMC iterations
B	Burn-in iterations
$\hat{R}$	Gelman–Rubin potential scale reduction factor

A. Notation

For the reader's convenience, Table 10 collects the principal symbols used throughout the paper. All vectors are set in bold.

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